

Western Neighborhoods Malley - Sébeillon, Lausanne, Switzerland

Climate Change and Energy Transition framework for
Sustainable Communities

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1 Introduction

The transition toward energy efficiency, driven by the need to decarbonize and embrace renewable energy sources, represents one of the most pressing challenges of our era. As a generation tasked with safeguarding the planet's future, we are navigating the dual imperatives of economic growth and sustainability. Urbanization is central to this transition, as cities have become the focal point of population growth, economic activity, and environmental impact. Currently, 55% of the world's population resides in urban areas, a figure that has grown from 30% in 1950 and is projected to reach 68% by 2050 (United Nations Department of Economic and Social Affairs [UN DESA], 2018). This demographic shift underscores that the future of humanity is decidedly urban.

Urbanization is both an opportunity and a challenge. Well-managed urban growth, guided by participatory planning and informed by long-term population trends, has the potential to maximize the benefits of agglomeration while minimizing environmental degradation and inequality. Sustainable Development Goal 11 commits to making cities inclusive, safe, resilient, and sustainable. However, unplanned urban expansion often leads to sprawl, pollution, and unsustainable consumption patterns. Urban areas, accounting for 71-76% of global CO₂ emissions and 67-76% of global energy use, exemplify the environmental strain that rapid urbanization can impose (Seto et al., 2014; UN-Habitat, 2011).

This report focuses on the **critical role of local actions in addressing these challenges**, recognizing that sustainability efforts must be tailored to the unique needs of each urban area. Switzerland, with its decentralized governance model, offers an example of how local policies can effectively balance urban growth with sustainability goals.

Understanding how cities like Lausanne can tackle the interconnected challenges of urban growth, energy efficiency, and environmental sustainability is essential for building a resilient future. Through the effective use of local governance, participatory planning, and innovative strategies, urban areas can pave the way for a more sustainable and equitable future for generations to come.

2 Methodology

The report consists of three main conceptual parts:

1. Contextualization.

This section outlines a broad policy, focusing on energy, housing and public spaces. All together it will help to demonstrate how Switzerland and particularly Lausanne is aligning Local Actions with Global Agendas, i.e. which concrete steps it takes to tackle challenges of decarbonisation and climate change adaptation.

2. Urban analysis

The second part focuses on the selected area of the Western Neighborhoods, providing an in-depth study and a rationale for its selection. This choice is grounded in the policy revisions of the city of Lausanne, along with its strategic plans and vision for urban development. The methods applied for the urban analysis include: situational analysis and diagnosis of the selected area, based on diverse data collection methods such as field studies, statistical data, and GIS analysis.

The GIS analysis of the selected area supports the sustainability assessment across key domains, including land use, housing stock, social infrastructure, mobility, transport, urban climate, energy potential, and vegetation.

3. Urban Redevelopment Strategy

Based on the evaluation of baseline and target KPIs, this section proposes a set of actions for the Western neighborhoods of Malley-Sébeillon. These actions address key measures across domains of Social, Architectural & Engineering Design, Urban Design, Sustainable Landscape Design Economic Design, Policy and Governance Design.

The proposed actions and measures aim to provide a holistic and comprehensive strategy for transforming the underdeveloped Malley-Sébeillon neighborhood into vibrant, sustainable, and inclusive communities, fully aligned with the principles of an ecoquartier. Furthermore, an evaluation of the co-benefits arising from the proposed strategy is included, emphasizing its potential positive impacts on the social, environmental, and economic dimensions of the area.

4. Implementation strategy

Serves as an orientation for local government, establishing feasible horizons of realization with comprehensive stakeholder involvement.

5. Conclusion

An overview of proposed strategy, resumes its main contributions and at the same time outlines methodological limitations.

3 Context | Policy revision

Global

National

Regional

City

3.1 Global

The global policy framework for sustainable urban development is anchored in a range of international agreements and guidelines. These emphasize the interconnected nature of housing, public spaces, and energy systems, aiming to create inclusive, resilient, and sustainable cities.

The 2030 Agenda for Sustainable Development

The 2030 Agenda, particularly SDG 11, serves as a roadmap for cities worldwide to address urban sustainability. Key targets include:

- *Target 11.1 - Access to adequate, safe, and affordable housing*
- *Target 11.7 - Developing inclusive green and public spaces*
- *11.b - Encouraging integrated policies to mitigate environmental and social risks.*

Achieving these targets requires integrating energy-efficient solutions, reducing emissions, and fostering climate adaptation into urban planning. (United Nations, 2015).

The New Urban Agenda (NUA), UN Habitat III, 2016)

Adopted at the UN Habitat III conference, the NUA complements the SDGs by providing a detailed action plan for sustainable cities, through putting emphasis on:

- Compact, inclusive urban development - Encouraging urban densification while maintaining access to green spaces to reduce sprawl and promote social cohesion.
- Energy efficient housing and infrastructure - Recommending policies to retrofit

existing buildings and adopt renewable energy sources.

- Participatory governance – Engaging local stakeholders in decision-making processes.

The NUA underscores affordability and resilience in housing through integrated energy systems and sustainable materials, whilst public spaces are a nexus for addressing mobility, energy consumption, and biodiversity goals (United Nations,2016).

UNECE Framework on Sustainable Cities

The UNECE Geneva UN Charter on Sustainable Housing provides a foundation for integrating housing and energy policies. It emphasizes:

- *Energy-efficient retrofits* - Reduce energy consumption within the existing housing stock
- *Cross-sectoral synergies* – Emphasize the alignment of housing policies **with urban energy systems**
- *Data-driven decision-making* - Leverage data for progress and inform evidence-based urban development.

The UNECE framework offers a roadmap for local governments to align their initiatives with international sustainability objectives. UNECE and ITU launched the United for Smart Sustainable Cities (U4SSC), an initiative that responds to SDG 11. Sustainable Cities. It provides tools such as the Smart Sustainable Cities Profiles (SSCPs), which assess city performance across three key dimensions: economy, environment, and society&culture, utilizing a mixed-methods approach that includes KPIs, along with surveys, interviews, and expert research. (UNECE, 2013, 2017, 2022)

Switzerland’s federalist governance structure divides responsibilities among federal, cantonal, and municipal levels. The federal government sets overarching objectives and legislation, while cantons implement and adapt these policies to regional contexts. This section highlights key policies influencing Sébeillon and Malley’s development and identifies gaps for further exploration.

Federal Energy Policies

Switzerland’s Constitution mandates sustainable development, with DETEC overseeing energy policies, including generation, efficiency, and mobility. Currently, 70% of energy is imported, with hydropower as the primary domestic source, but oil and gas still comprise 49% of consumption, emphasizing the need for a renewable transition (Presence Switzerland, n.d).

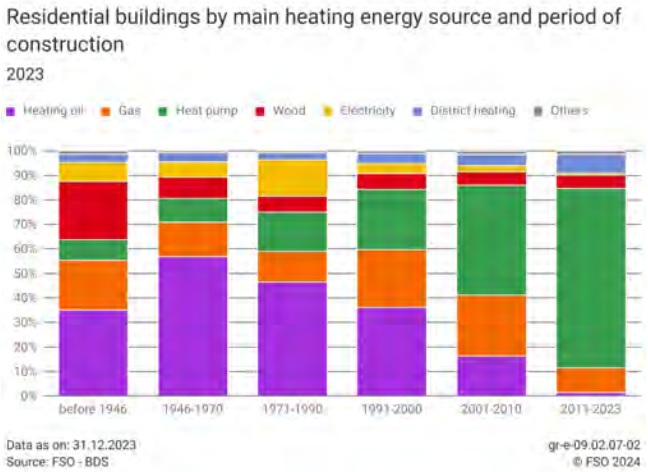


Fig. 1 - (Source: FSO, 2024)

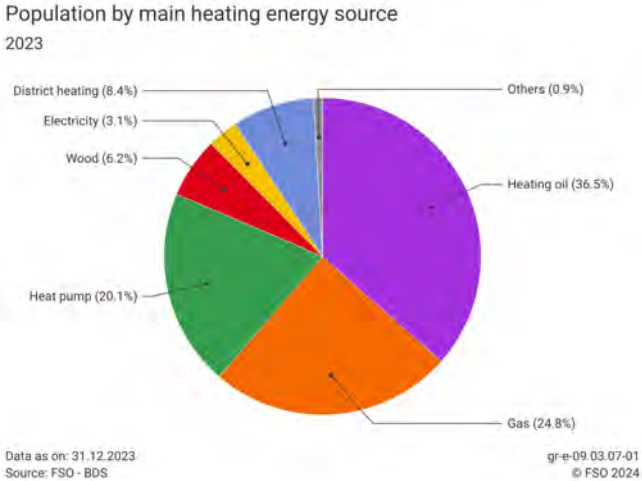


Fig. 2 - (Source: FSO, 2024)

Following the 2011 Fukushima incident, Switzerland initiated a nuclear phaseout and adopted the Energy Strategy 2050, targeting a 53% reduction in per capita energy consumption by 2050 compared to 2000 levels (The Federal Council, 2012). Policies focus on energy efficiency, particularly in new build construction and industrial production, as one-third of energy use goes to winter heating. The Swiss Buildings Programme enforces stricter insulation and renewable heating standards while promoting on-site renewable energy production in buildings.

To fund the transition, CO2 levies (CHF 120/tonne) and the Renewable Energy Promotion Contribution (2.3 cents/kWh) finance renovations and renewable projects. By 2050, 45,000 GWh of electricity is expected from renewables and 39,200 GWh from hydropower. A CHF 1.7 billion fund supports grid modernization. (Swiss Federal Office for the Environment, 2024; Swiss Federal Office for the Environment, 2017).

Despite progress, the energy transition re-

mains slow due to high investment costs. Switzerland is expediting infrastructure licensing to meet net-zero targets by 2050. Continued federal and cantonal oversight is essential for achieving a resilient and sustainable energy future.

Federal policies for Sustainable Development & Housing

Switzerland aligns its sustainability policies with global goals, emphasizing economic growth, environmental responsibility, and social equity (Swiss Federal Office for Spatial Development, 2011). The Spatial Planning Act limits urban sprawl by separating building zones ad non-building zones, while the Swiss Biodiversity Strategy mandates ecological corridors to connect habitats and enhance biodiversity (Swiss Federal Office for Spatial Development, 2011; Swiss Federal Office for the Environment, n.d.) .

Urban planning efforts focus on integrating greenspaces into cities, blending wild and urban environments to promote ecosystem health. Compact, mixed-use developments near public transport hubs are prioritized to reduce resource consumption and improve urban liveability (Swiss Federal Office for Spatial Development, 2011).

Sustainable transport policies emphasize cycling, pedestrian infrastructure, and public transit to create accessible, low-emission cities. Additionally, experiments with circular economy models support resource efficiency and sustainable development (Swiss Federal Office for Spatial Development, 2011).

3.3 Regional

The Canton of Vaud, the third most populous canton in Switzerland with approximately 846,300 residents (Population État de Vaud, 2023), has outlined its urban development strategy through the Agenda Vaudois 2030 and the Plan Directeur Cantonal Vaud 2050. These documents prioritize ecological, economic, and social goals, aiming for carbon neutrality by 2050.

The Agenda Vaudois, established in 2003, integrates sustainability into public policies across various sectors. During its 2017-2022 legislative program, the Council of State aligned with the UN 2030 Agenda, reinforcing its commitment to a sustainable future (OCDC, 2021). In July 2023, the Plan Directeur Cantonal Vaud 2050 was adopted, providing a long-term framework to enhance climate resilience and address seven critical urban development challenges (PDCn 2050, 2023):

- 1. **Natural environment** emphasizes the need to preserve biodiversity and ecological systems, ensuring that urban development does not hinder natural habitats.
- 2. **Agriculture** emphasizes sustainable land use and food security by protecting agricultural land crucial for local food production amidst urbanization pressures. Policies aim to balance urban growth with fertile land preservation and enhance resilience to environmental challenges.
- 3. **Built Spaces** focus on preserving historical architecture while promoting modern, resilient structures. High-density housing is encouraged to limit urban sprawl and optimize land use.
- 4. **Services** seek to enhance public facilities like healthcare, education, and social in-

frastructure. The goal is equitable access to essential services in both urban and rural areas, enhancing quality of life.

- 5. **Mobility** advocates for sustainable transportation by investing in pedestrian and cycling infrastructure and improving public transit. These measures aim to reduce congestion and CO2 emissions.
- 6. **Economic Activities** prioritize innovation and economic growth by supporting start-ups, research institutions, and industrial diversification, fostering a competitive and resilient economy.
- 7. **Resources and Energy** promote renewable energy use and resource efficiency. Policies support energy-efficient buildings, circular economy practices, and sustainable water management.

These strategic initiatives position the Canton of Vaud toward a sustainable and resilient future. Achieving net-zero by 2050 requires a significant shift in energy practices. The canton has committed to accelerating its energy transition to eliminate fossil fuels, which account for 67% of its greenhouse gas emissions (Statistique Vaud, 2023).

A prospective analysis outlines scenarios for reducing fossil fuel consumption: Scenario 1 projects a 42% reduction, Scenario 2 a 68% reduction, and Scenario 3 an 83% reduction. These reductions, coupled with renewable energy development, would enhance the canton's energy independence (Statistique Vaud, 2023).

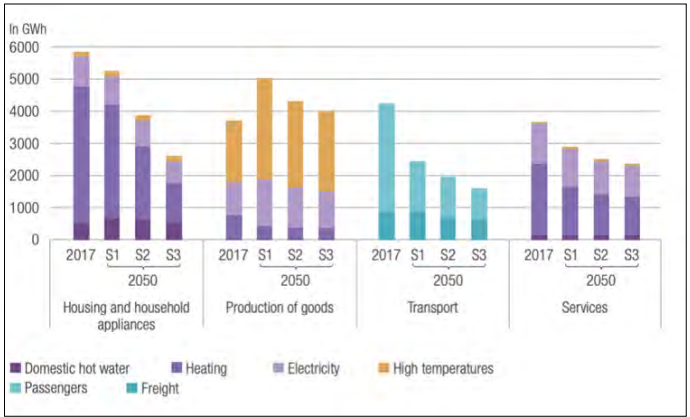


Fig. 3 - Final energy consumption by sector in Vaud in 2017 and projections for 2050 (Source: Statistique Vaud, 2023)

Additionally, a model-based analysis from the **Plan Directeur Cantonal Vaud 2050**, published in June 2024, highlights the temperature differences between green spaces and dense urban areas on a summer day. This underscores the importance of implementing proposed measures, as failing to do so could result in significant environmental consequences (*Situation climatique du Canton de Vaud – PDC 2050*, 2024).

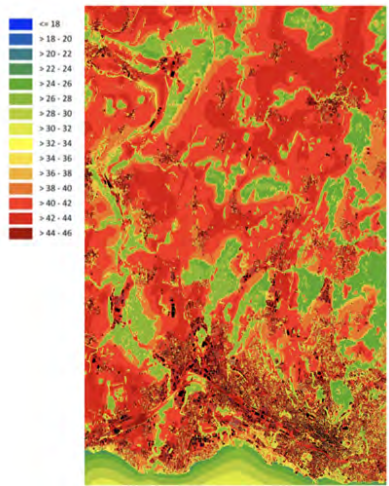


Fig. 4 - Temperature Projected in 2060 – Lausanne, Vaud (Source: *Situation climatique du Canton de Vaud – PDC 2050*, 2024)

3.4 City | Lausanne’s Energy, Housing and Public Space Policy

Lausanne has a long track record of sustainable development, being a global leader in energy efficiency, mobility, city planning, as well as educational and social affairs. The city has held the Energy City label since 1996 and achieved the European Energy Award GOLD label in 2004 (European Energy Award, 2014). As a signatory to the Aalborg Charter, Lausanne’s local involvement is reflected in the introduction of Agenda 21, an action plan designed to achieve a high-quality environment, an inclusive society, a healthy economy, and sound finances. The city is also noted for its comprehensive urban transport system, including a full metro network for a population of around 140,000, highlighting its dedication to climate adaptation, mitigation, and sustainable development goals (Ville de Lausanne, 2020).

Building on Federal and Cantonal policies surrounding energy and housing, Lausanne is at a critical juncture, aiming to support a growing population while striving for carbon neutrality by 2050. The city is implementing measures that go beyond federal and cantonal actions to improve energy efficiency and address climate change (Ville de Lausanne, 2020).

Energy Policy

By aligning with Federal and Cantonal energy efficiency policies, Lausanne’s city-wide energy policies aim to achieve carbon neutrality by 2050. Key measures include the strategic expansion and densification of district heating systems, improving building insulation, and transitioning to 100% renewable heat by 2035 with 75% energy self-sufficiency by 2050 (Ville de Lausanne, 2020).

Lausanne has adopted the Equiwatt plan to meet federal and local renewable energy goals by providing an array of subsidies for the purchase of greener technologies for all social classes, such as solar PVs and e-bikes (Ville de Lausanne - Equiwatt, n.d.). By coupling zero-emissions goals with social concerns, such as population growth, the City of Lausanne is working on synergies that enable the city to grow while protecting its historical heritage and greenspaces. Therefore, the city is leveraging its unique characteristics along the lake and situated between Geneva and the Alps.

Housing and Public Space Policy

Like many Swiss cities, Lausanne has a low homeownership rate of under 40% (Ville de Lausanne, 2024), resulting in a tight housing market and high rents. As of January 2025, the median rent was CHF 347 per square meter, equivalent to around CHF 2,292 for a 3.5-room apartment (RealAdvisor, 2025). To address housing precarity, Lausanne has implemented rent control measures and subsidized housing programs, offering apartments at 20% to 40% below market rates. These programs are accessible to eligible individuals, including foreign nationals with specific residency permits, and aim to create a more inclusive housing landscape. Despite these efforts, the city still faces housing shortages, with a vacancy rate of just 0.96% (État de Vaud – Statistique Vaud, 2024).

Lausanne is also advancing renewable energy and urban development projects to promote sustainability. Initiatives include the installation of 24 wind turbines by Eolien Romande SA, capable of generating 110 GWh annually for



Credit: <https://switzerlandtravelcentre.com/fr/che/offre/lausanne-ville-gourmande-sur-le-lac-leman>

30,000 households, and deep geothermal systems to supply sustainable heating (SI-REN, n.d.). Solar energy plays a prominent role, with photovoltaic systems across municipal buildings aiming to generate 50 GWh annually to meet the needs of 14,000 homes (SI-REN, n.d.).

These energy efforts are integrated into urban development projects like the Plaines-du-Loup eco-district, a model for harmonizing energy efficiency, urban greening, and housing. This project transforms post-industrial land in western Lausanne while preserving historical districts and protecting arable land. As part of the Projet Métamorphose, the eco-district includes 30% subsidized housing, 40% capped-rent housing, and 30% free-market housing, fostering social diversity and equity. The development aligns with the 2,000-watt Society vision, incorporating energy efficiency measures and connecting to public transport expansion, including the future m3 metro line (Ville de Lausanne – Plaines-du-Loup, n.d.).

Lausanne also prioritizes health and public spaces as key elements of its sustainability strategy. The “One Tree, One Child” initiative has planted a tree annually for every newborn since 2001, enhancing green spaces and raising environmental awareness. The city has implemented over 35 measures to improve access to safe public spaces, with a strong focus on equity and social cohesion, aligning with SDG 11 to foster an inclusive and sustainable urban environment (Forbat et al., 2024).

Integration of Energy and Housing Policies

Lausanne’s energy and housing policies under

the Plan Climat (Climate Plan) are quite extensive, highlighting four key aspects:

- 1. Carbon neutrality goals** (zero direct emissions in mobility by 2030, zero direct emissions across all sectors by 2050)
 - a. Improving energy efficiency through heating improvements (building retrofits, expanding and densifying district/centralized heating, replacing oil and gas heating systems with combined heat and power plants [CHP], combined cycle gas-fired power plants).
 - b. Increasing the use of renewable energy (solar thermal heating and PVs, geothermal, biomass, thermal exchange with Lac Lemman)

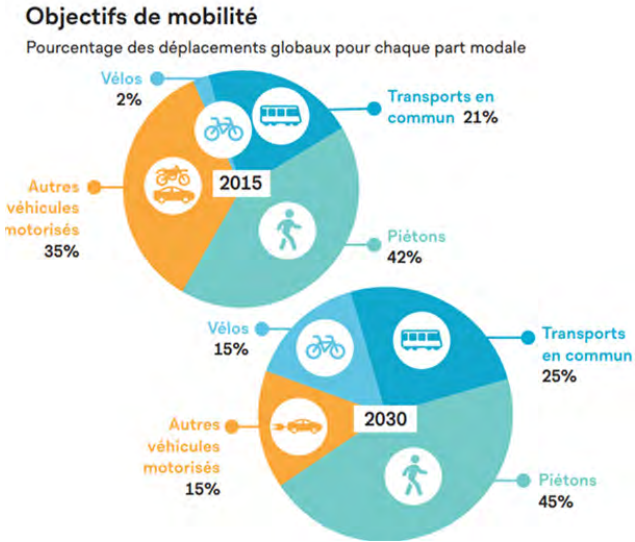


Fig. 5 – Mobility objectives under the Plan Climat, Percentage of overall trips for each modal share (Source: Ville de Lausanne - Plan Climat Brochure, 2022).

2. Sustainable mobility goals (eliminate combustion engine cars by 2030)

- a. Reducing the use of owner-operated vehicles by improving and expanding mobility networks (bike transit infrastructure and public transport options, car-free zones, multimodal transport options, removing parking for green-spaces, Equiwatt e-bike subsidies).
- b. Building the electrification capacity of the grid (solar PVs on rooftops and facades - Equiwatt, vehicle to grid technology, energy storage).

3. Urban adaptation goals

- a. Reduce heatwaves and encourage a cool city (increase urban green space by 50% by 2040, doubling the tree canopy index, vegetated roofs for insulation and greenery, passive cooling techniques, thermal exchange with Lac Lemán).
- b. Manage flood and drought risks (green spaces and soil permeability, encourage rainwater harvesting, expand the blue-green infrastructure, modernize stormwater systems).
- c. Enhancing biodiversity (expand green and blue corridors, preserving and expanding forests, promoting biodiversity focused landscaping).

4. Social equity

- a. Costs and benefits of sustainability policies are distributed fairly (free public transport for people receiving social benefits, discounted fares for other vulnerable populations, rent protection against green gentrification, aid for retrofit and energy costs, development of community gardens, local cooling facilities).

Such comprehensive policies were selected to ensure co-benefits, for example encouraging bike use through infrastructure development and e-bike subsidies reduces CO2 emissions while improving air quality and healthy citizens. Additionally, Lausanne's approach to climate policy is highly participatory. For example, the "Cool Spaces Map" allows residents to share locations offering cool refuges in summer, both indoors and outdoors. This initiative encourages reduced air conditioning use, promotes shared cooling spaces, and fosters a sense of community and belonging, critical in a rapidly expanding and diversifying city like Lausanne (Plan Climat - Ville de Lausanne, 2022).



Fig. 6 - Carte Interactive des Espaces Frais "Cool Spaces Map" Source: (Plan Climat - Ville de Lausanne, 2022).

Summary

Lausanne's policies strongly align with global, federal, and cantonal sustainability frameworks, emphasizing housing affordability, renewable energy, and urban greening. The city's initiatives reflect the UN's 2030 Agenda and the New Urban Agenda by promoting compact, energy-efficient developments and participatory governance, such as the "Cool Space Map". Lausanne also supports Switzerland's Energy Strategy 2050 with plans for carbon neutrality, including district heating and renewable energy integration.

These alignments underscore the city's commitment to reducing emissions, improving resilience, and fostering inclusivity. However, challenges remain. Despite progressive housing policies, a low vacancy rate of 0.96% highlights ongoing supply issues, and scaling up renewable energy infrastructure is slow due to high costs. While urban greening plans, like a 50% increase in green space (Plan Climat, 2023), are ambitious, their implementation needs acceleration to mitigate climate risks effectively. Social equity measures, including subsidized housing and protections against displacement, are robust but require expansion to address rising rents and population growth. Overall, Lausanne's efforts are commendable, yet addressing gaps in housing, energy scalability, and climate adaptation will be essential to achieving its long-term sustainability goals.

Data Accessibility

The Swiss Federal Statistical Bureau offers an extensive database at both national and regional levels, covering key areas like the economy, households, incomes, population, and energy consumption. Similarly, the Lausanne Statisti-

cal Bureau provides detailed, city-specific data essential for addressing local urban issues.

However, while these databases are comprehensive, they often lack the granularity needed for in-depth analysis, or access to detailed datasets may be restricted. Geospatial data repositories are hosted on three primary government platforms: the federal repository (map.geo.admin.ch), the cantonal repository for Canton Vaud (geo.vd.ch), and the municipal platform (map.lausanne.ch). These platforms provide datasets on land use, infrastructure, and environmental features. However, many datasets are limited to raster formats rather than vector formats, reducing their usability. High-resolution urban data, such as building heights and typologies, is often missing or must be sourced from platforms like OpenStreetMap, which may lack the accuracy required for detailed urban analysis.

Data accessibility for sustainability assessments in Malley-Sébeillon is quite extensive though at the beginning can look fragmented and unevenly distributed. There is also an uneven distribution of open-access data that limits the potential for conducting holistic and effective sustainability assessments.

To bridge these gaps, improvements in transparency, interoperability, and open access to critical datasets are essential. Specific focus areas include detailed urban morphology datasets. These improvements are critical for enabling data-driven, sustainable development strategies. This inconsistency creates a significant barrier for research.

4 Urban Redevelopment | Analysis

Malley - Sébeillon Neighborhoods —

4.1 Context| Malley - Sébeillon

Total area 46,12 he	Population Density - 114 hab/per hectare
Current Land Use:	Vacancy rate - 0,67%
• 26, 2 % Residential (12,1 he)	Green Space inside current land use - 19, 24% (without parks)
• 3,9 % Commercial (1,8 he)	Parks - 20,1he
• 41, 4 % Industrial (19,11 he)	
• 21, 2 % Administrative (9,8 he)	
• 7,17 % Transport (3,31 he)	

Figure 5. Western Neighborhoods.

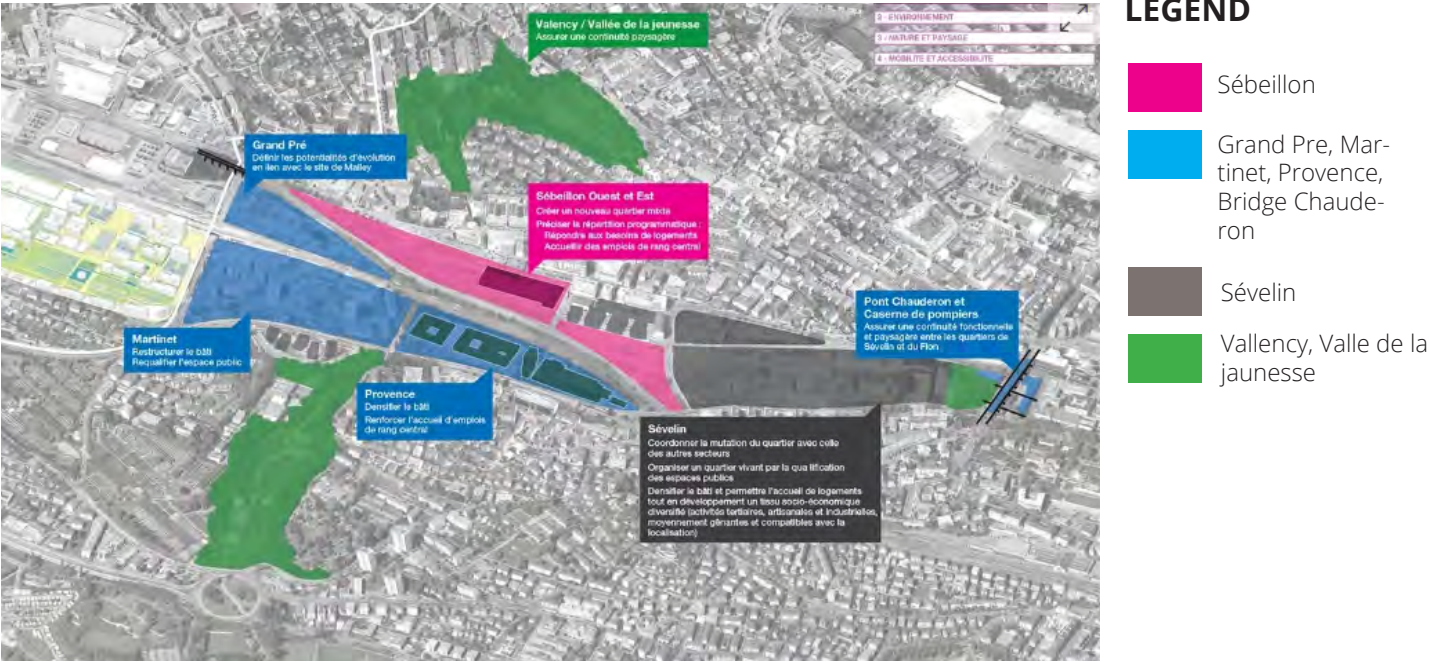


Source: Ville de Lausanne, 2024.

According to Swiss Sustainable Development Strategy (SWISS, 2024) the neighborhood scale is particularly appropriate for implementing concrete operational actions to achieve sustainable development. It is emphasized that at this scale, urban realities can be addressed comprehensively—large enough to engage with sustainability criteria beyond the scope of a single building, yet sufficiently defined to enable clear visualization and execution of specific interventions. Along with this, this scale allows for tangible applications of sustainable development, including creation and enhancement of accessible, inclusive public spaces, construction of ecological and energy-autonomous structures, promotion of social and intergenerational mix through housing and community initiatives and development of infrastructure that promote active mobility, such

as pedestrian and cycling pathways. By integrating these elements, a project of sustainable neighborhood not only addresses sustainability at an operational level but also ensures that urban transformations contribute to a high standard of living and a more connected, resilient city. In order to prioritise urban densification as one of the central focuses of the current work, the scale of several interconnected neighborhoods was chosen. This choice aims to close one «mental» gap in Lausanne’s urban fabric and harmonize the central area of the city, creating a cohesive and well-integrated urban environment. As part of the city’s strategic development program «Grands Projets», there exist clearly defined objectives set forth by the municipality.

Figure 6. Parcels Malley-Sébeillon for Grands Projets.



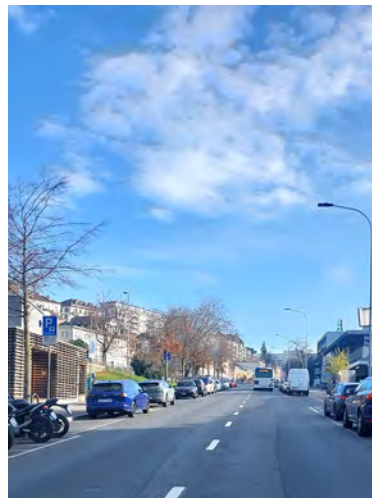
Source: Ville de Lausanne, 2024.

- They are:
1. Define connectivity with adjacent quarters.
 2. Restructure existing buildings.
 3. Requalify public spaces.
 4. Densify the urban fabric.
 5. Strengthen the reception of central-ranking employees.
 6. Ensure landscape and functional continuity between neighboring quarters.
 7. Develop a new mixed-use quarter.
 8. Specify programmatic distribution for mixed-use development.
 9. Address housing needs with balanced residential and economic uses.
 10. Maintain and enhance landscape continuity.
 11. Coordinate the transformation of quarters with developments in adjacent sectors.
 12. Organize vibrant quarters with qualified public spaces.
 13. Allow the reception of housing within a diversified socio-economic fabric.
 14. Develop artisanal, tertiary, and industrial activities compatible with urban settings.

It is also worth mentioning that the participatory planning process is already well underway, the main results of which were summarized on the official site of the administration (see the annex).

4.2 Field study

The area evokes mixed feelings. On one hand, it is full of potential, with charming community-driven initiatives such as communal gardens and cozy pocket parks in the residential courtyards of Sévelin. Public spaces in this area are lively, with young people walking around the Malley neighborhood, and street retail along Avenue du Chablais is offering a variety of cafes, pharmacies, and other services. On the other hand, it is difficult to identify the area as a cohesive whole. The presence of a large



Credits: authors

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highway along Avenue Jules Gonin dominates the landscape, and the roads are heavily car-oriented. There a lot of unused parking, all this creates an environment where walking doesn't feel entirely safe, even in seemingly pedestrian-friendly but heavily overloaded with warehouses zones like Sévelin, home to landmarks such as the Sévelin theater, Café Datcha, and the concert hall, which are well-known throughout the city.

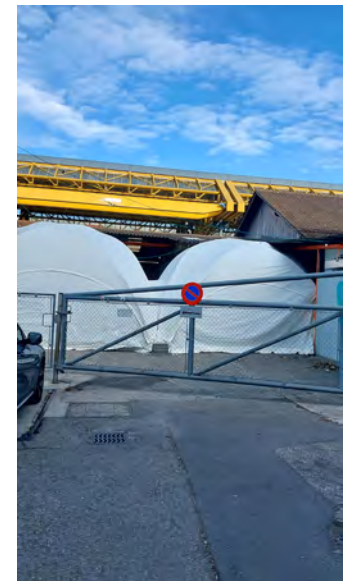


Figure 7. Mental map. Based on «Image of the city» (K.Lynch)*



Source: authors

A mental map is a map recreated from memory by a pedestrian to represent their understanding of a specific place. These maps often highlight objects or landmarks that are remembered for their distinctive features (e.g. paths, edges, districts, nodes, and landmarks) or are associated with subjective perceptions of the space and emotional connections or memories. Those distinctive components help individuals navigate and structure their understanding of a place.

- «mental» stops for walking (not safe, nothing happens, no exit)
- itinerary of field studies
- ~ mental borders separating one area from another

4.3 Socio-Economic Dynamics | Demographics

Figure 8. Basic Demographic profile

Aperçu 2016		Quartier 3		Lausanne	
Habitants	11'269	100.0%	143'561	100.0%	
dont étrangers	5'801	51.5%	61'581	42.9%	
dont nés en Suisse	1'066	18.4%	10'187	16.5%	
dont permis C	3'085	53.2%	32'262	52.4%	
dont Suisses	5'468	48.5%	81'980	57.1%	
dont nés en Suisse	4'058	74.2%	64'366	78.5%	

Aperçu 1980		Quartier 3		Lausanne	
Habitants	8'352	100.0%	128'571	100.0%	
dont étrangers	2'450	29.3%	30'217	23.5%	
dont permis C	1'841	75.1%	20'370	67.4%	
dont Suisses	5'902	70.7%	98'354	76.5%	

Ménages 2016		Quartier 3		Lausanne	
1 personne	2'354	44.0%	34'027	47.8%	
2 personnes	1'500	28.0%	19'326	27.2%	
3 personnes	726	13.6%	8'674	12.2%	
4 personnes	516	9.6%	6'361	8.9%	
5 personnes et plus	252	4.7%	2'787	3.9%	
Total	5'348	100.0%	71'175	100.0%	

Source: Ville de Lausanne, Contrôle des habitants et Guichet cartographique, 2024.

Quartier 3* is home to **11,833 residents (2023)**, comprising 5,904 Swiss nationals (49.9%) and 5,929 foreign nationals (50.1%), primarily descendants of immigrants from Southern European countries (35,4%) who arrived in the late 1960s and 1970s. The area has a population density of approximately **114 inhabitants per hectare**, significantly higher than Lausanne's average urban density of 82 inhabitants per hectare.

The percentage of **older residents (aged 60 and above)** is relatively low at **1,282 individuals (6.1%)**, while the proportion of youth under 20 years old is similarly modest, **at 2,282 individuals (8.1%)**, including 101 children under 1 year old. There is also a prevalence of male population (53%) over female (47%). The average household size is 1 person, reflecting the general trend observed across Lausanne.

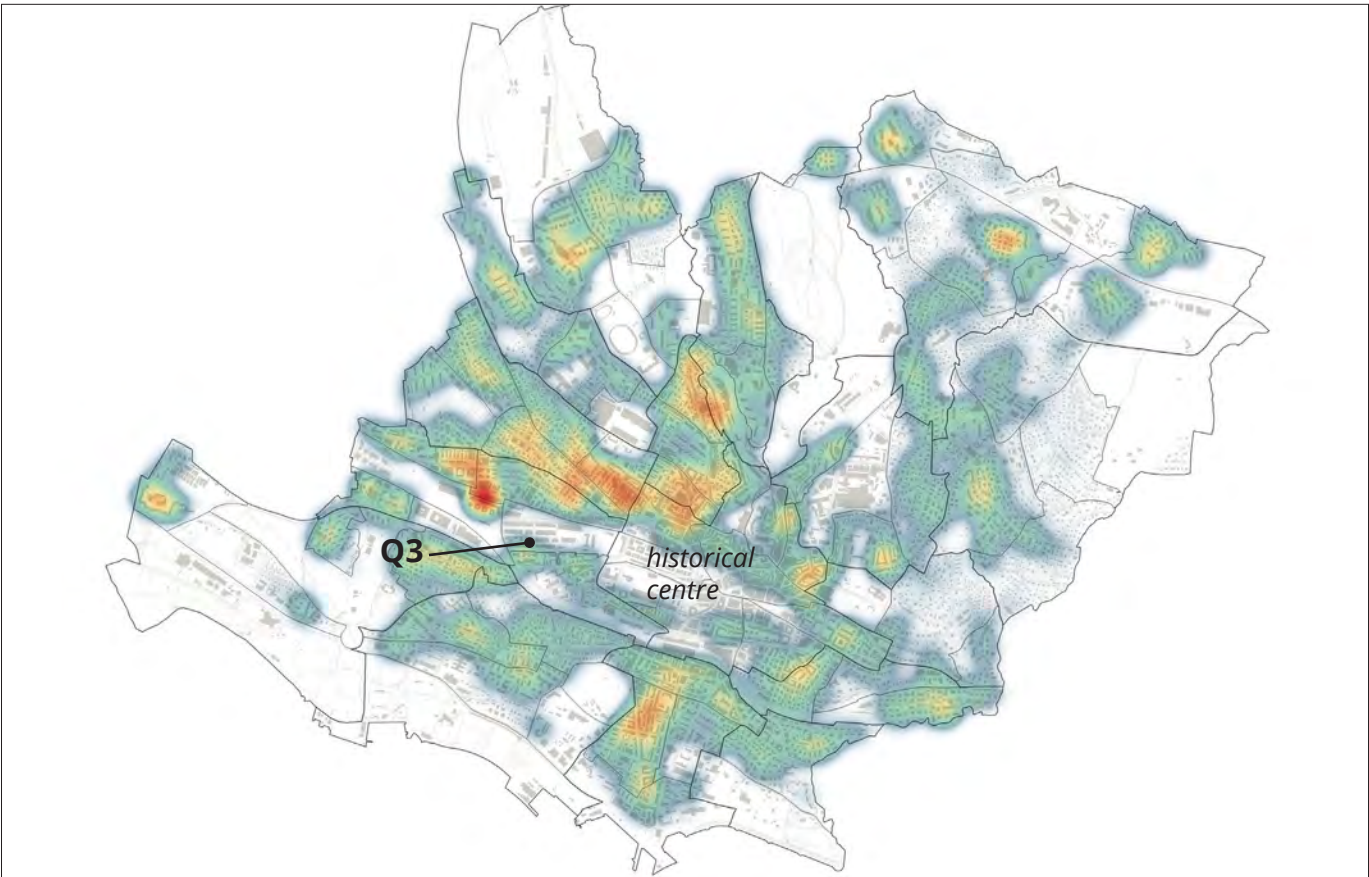
Figure 10. Demographic Division by Nationality

Groupes de pays 2016		Quartier 3		Lausanne	
A	Europe du Sud-Ouest	2'055	35.4%	21'680	35.2%
B	France	860	14.8%	11'440	18.6%
C	Europe du Sud-Est	839	14.5%	6'738	10.9%
D	Afrique subsaharienne	452	7.8%	4'107	6.7%
E	Europe du Nord	238	4.1%	4'009	6.5%
F	Amérique latine	349	6.0%	3'178	5.2%
G	Afrique du Nord et Moyen-Orient	308	5.3%	3'230	5.2%
H	Europe de l'Est et Asie centrale	222	3.8%	2'591	4.2%
I	Sous-continent indien	222	3.8%	1'451	2.4%
J	Asie de l'Est et Sud-Est	136	2.3%	1'751	2.8%
K	Amérique du Nord et Océanie	115	2.0%	1'319	2.1%
Total (y c. apatrides et inconnus)		5'801	100.0%	61'581	100.0%

Source: Ville de Lausanne, Contrôle des habitants et Guichet cartographique,

*The exact data on residents in the designated area of Malley-Sébeillon was not available from official sources.

Figure 9. Population Density in Lausanne



Source: Ville de Lausanne, Contrôle des habitants et Guichet cartographique, 2024.

Figure 11. Gender Distribution

Habitants du Quartier 3 par groupes de pays 2016													
Sexe	A		B		C		D		E		F		Etrangers
Homme	1'091	53%	501	58%	433	52%	229	51%	124	52%	166	48%	3'046 53%
Femme	964	47%	359	42%	406	48%	223	49%	114	48%	183	52%	2'755 47%

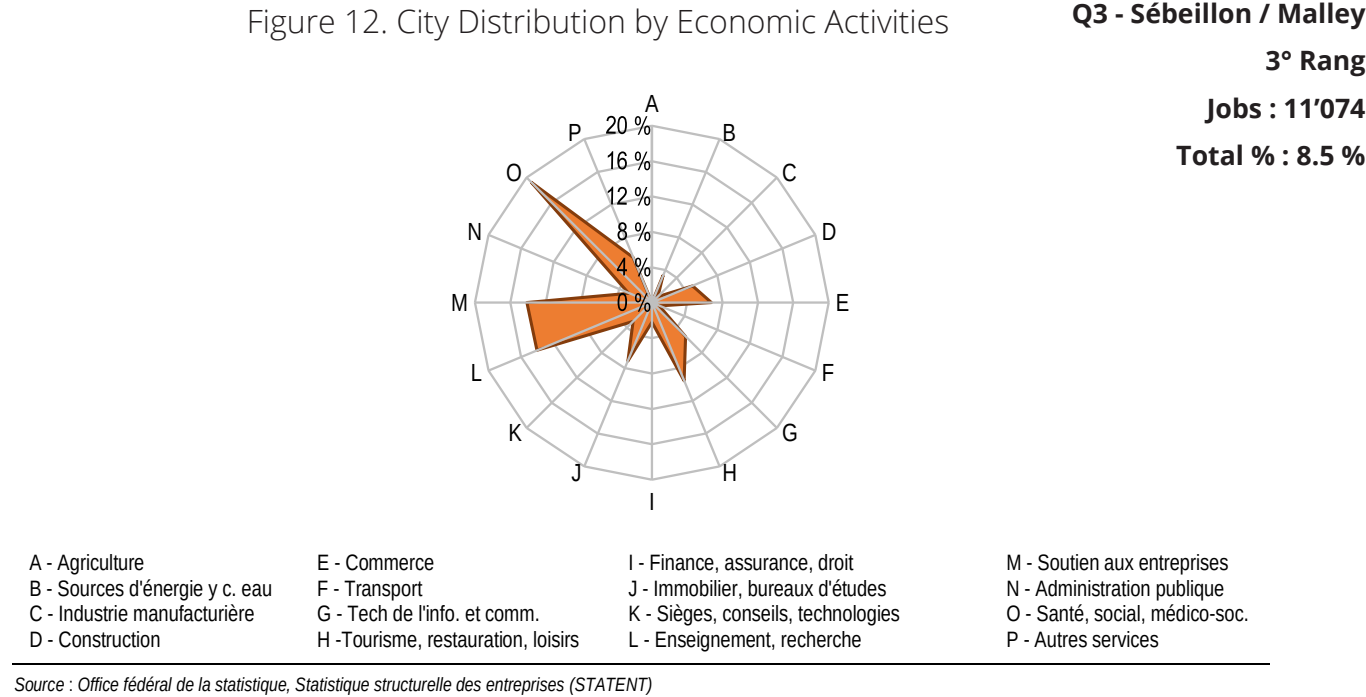
Âges	A		B		C		D		E		F		Etrangers
< 20 ans	385	19%	116	13%	201	24%	130	29%	30	13%	50	14%	1'134 20%
20 à 39 ans	699	34%	604	70%	301	36%	191	42%	149	63%	163	47%	2'650 46%
40 à 64 ans	761	37%	110	13%	292	35%	122	27%	44	18%	129	37%	1'673 29%
65 ans et +	210	10%	30	3%	45	5%	9	2%	15	6%	7	2%	344 6%

Total	2'055	100%	860	100%	839	100%	452	100%	238	100%	349	100%	5'801 100%
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Source: Ville de Lausanne, Contrôle des habitants et Guichet cartographique, 2024.

4.3. Socio-Economic Dynamics | Economic activities and job market

Figure 12. City Distribution by Economic Activities



Source: Ville de Lausanne, 2024

Q03 is home to **11,074 workplaces**, accounting for approximately **8% of Lausanne's total employment**, highlighting its significant role as an employment hub within the city. The Jobs Per Resident ratio in Q03 is among the highest in Lausanne, making it the **third most important concentration of workplaces** in the city. This underscores the district's strategic economic importance and its strong capacity to attract and sustain employment opportunities.

Around 80% of jobs are concentrated in the private sector, in areas such as finance, insurance, law, real estate, consulting firms, headquarters, advisory services, technology, business support, and other miscellaneous services.

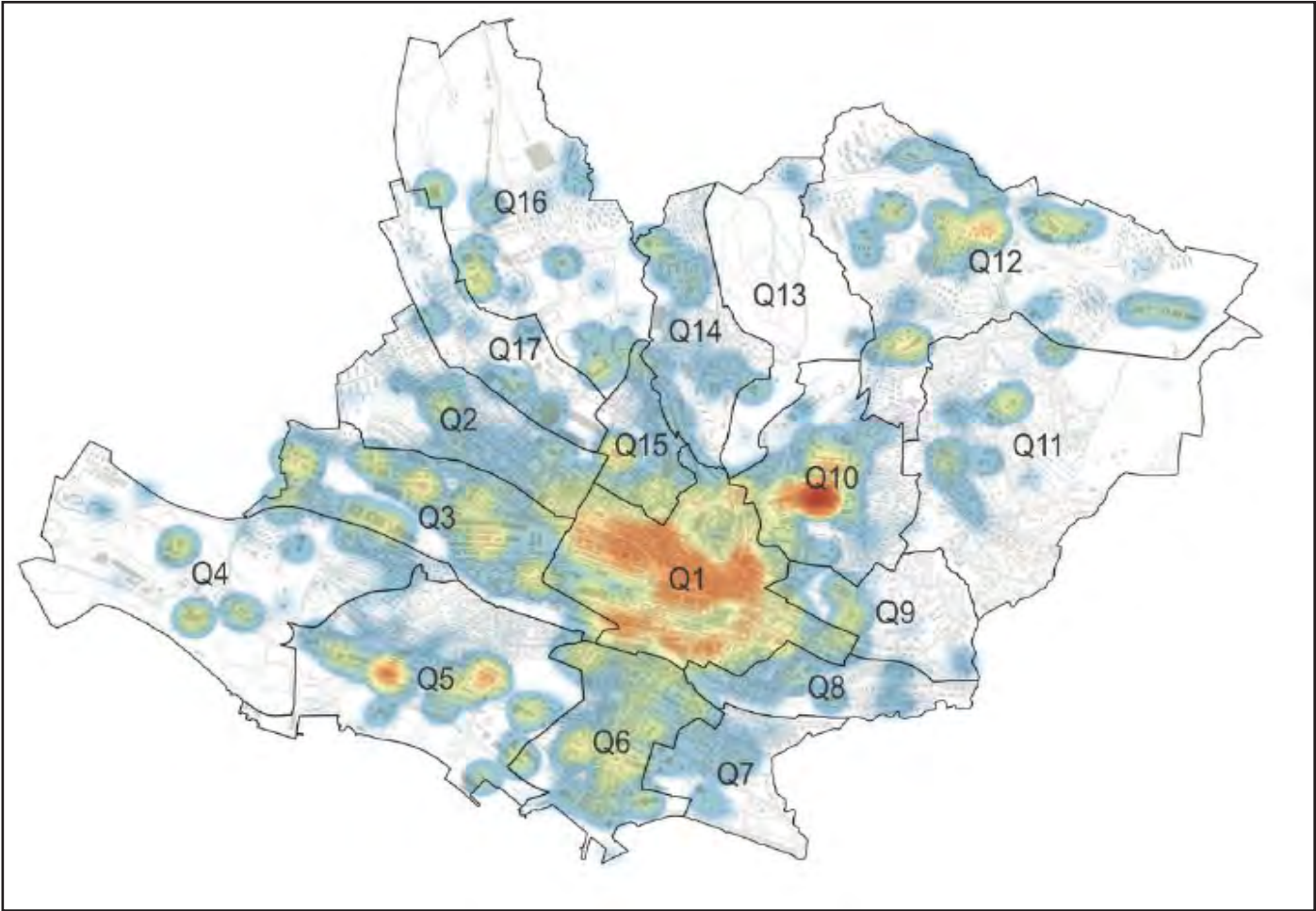
In addition, the district hosts a variety of diverse economic activities, with a notable presence in sectors such as manufacturing, construction, commerce, and transportation (see Figure 5 for details).

Figure 14. Jobs distribution by gender in 2022.

	Femmes	Hommes
1 - Centre	51.4 %	48.6 %
2 - Maupas / Valency	56.2 %	43.8 %
3 - Sébeillon / Malley	42.8 %	57.2 %

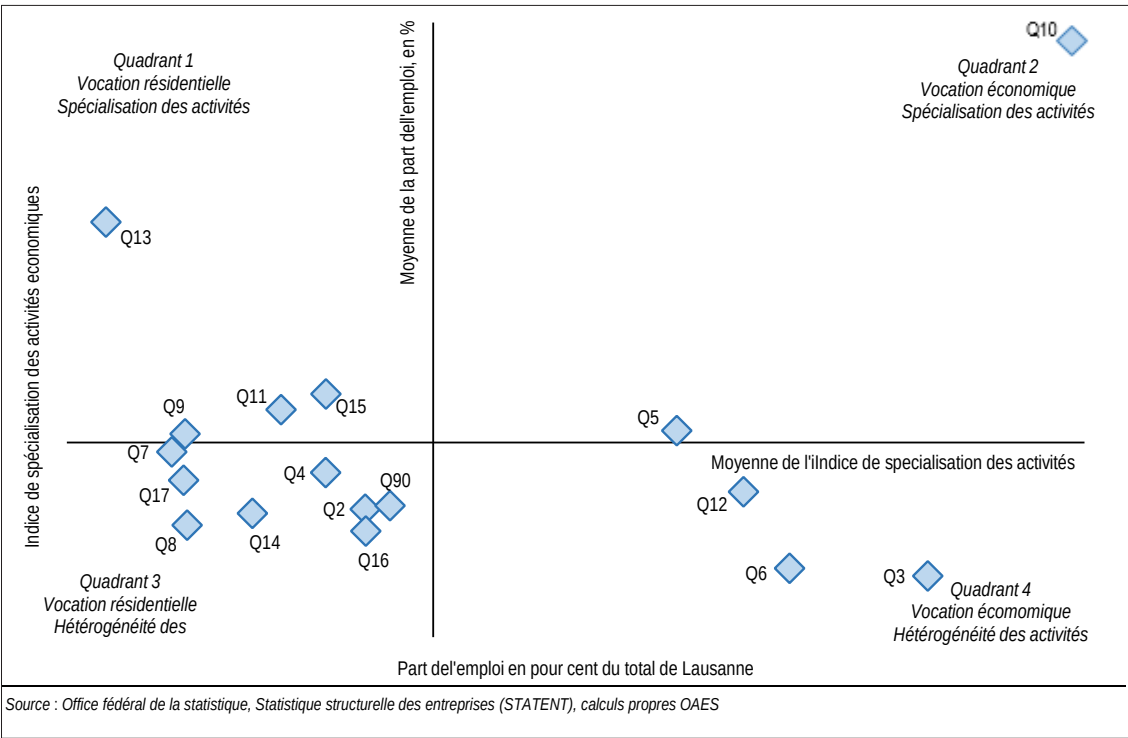
Source: Ville de Lausanne, 2024

Figure 13. Density of jobs



Source: Ville de Lausanne, 2024

Figure 15. The territorial vocation of neighborhoods.



Source : Office fédéral de la statistique, Statistique structurelle des entreprises (STATENT), calculs propres OAES

Strength - Central position in the city center - Close to public transport - Close to parks - Strong Municipality and Administration Support - Participory Practice of Planning - Presence of Established Commercial Infrastructure - Presence of cultural and community hubs	Weakness - Weak connectivity for soft mobility - Lack of identity - Industrial look - Limited pedestrian-friendly areas - Underutilized public spaces - Lack of green building infrastructure
Opportunity - Becoming a model for sustainable transition - Setting a benchmark for policy excellence - Leveraging social diversity and industrial heritage - Potential for reducing housing scarcity with affordable housing options - Reduction of negative externalities by enhancement of pedestrian and cycling infrastructure - Attracting green and tech-focused businesses - Improving connectivity with surrounding districts and boost economic activity to a new level	Threats - The area may remain marginal - Risk of becoming overly corporate, displacing community - Increased property values may lead to gentrification - Resistance to change from local businesses or residents - Potential loss of industrial heritage during redevelopment - Environmental challenges in redevelopment projects - Lack of coordination between public and private stakeholders

Source: authors
Challenges and Transformations Malley-Sébeillon

The district faces challenges typical of areas transitioning from industrial use to mixed urban functions, including integrating residential spaces, enhancing social inclusion, and preserving cultural heritage. Ongoing projects reflect efforts to address these challenges by fostering community engagement.

Key challenges: - Limited Connectivity: Poor integration with adjacent areas. - Marginalized Appearance: Industrial and fragmented visual identity. - Insufficient Landscape Connectivity: Lack of cohesive green corridors. - Low Urban Density: Underutilized land and spaces. - Integration and Cohesion: Disjointed urban fabric.	
KPI development	
Objectives	KPI Category
Rebalance land use to prioritize mixed-use and residential zones	Land Use & Density
Increase housing supply	Housing
Expand and enhance green spaces	Environment
Deepen the integration of soft mobility solutions by improving pedestrian and bike network	Mobility
Strengthen and stimulate the local economy	Economy, social
Enhance social diversity and inclusivity	Social
Increase and optimize local renewable energy production	Environment
Increase of Public Realm	Social

Source: authors

Objective

Transform an underdeveloped brownfield parcel into a productive, sustainable, and inclusive urban space through strategic interventions and data-driven KPIs.

4 Urban analysis |

4.5 Situational Analysis & Diagnosis

4.5.2 Sustainability assessment | Key Performance Indicators (KPI)

According to the criteria for eco-neighborhoods (SWISS, 2024), which encompass categories such as density, functional mix, sustainable mobility, high environmental quality, and intergenerational and social diversity, a set of Key Performance Indicators (KPIs) based on these criterias has been developed to establish a baseline for assessing the sustainability potential of Malley-Sébeillon. However, due to limited

Transport&Mobility

KPI	Target Value
Public Transport Network Coverage	100% of the area within 500m of a transport stop
Public Transport Network Access	90% of residents within 300m of a public transport stop
Pedestrian infrastructure	100% of sidewalks meeting accessibility & safety standards
Walkability score	Minimum score of 80/100

Economy

KPI	Target Value
% of Non-Residential Activities	30% of urban space dedicated to commercial activities
Number of Jobs per Resident	0.8 jobs per resident

Environment

KPI	Target Value
Noise Exposure	Below 55 dB in residential areas
Green Areas	Minimum 10 m² per resident
Green Area Accessibility	100% of residents within 300m of a green space
Pervious surfaces	Minimum 40% of total land area
Urban Tree Canopy Cover	25% of total area

data available from open sources, some indicators will be proposed as targets for future assessment, based on recommendations to be provided later. The methodological framework for the KPIs was adopted from the UNECE Framework on Sustainable Cities.

Social

KPI	Target Value
Public Space Ratio	Minimum 20% of total land area
Recreational facilities	1 playground per 2,000 residents
Access to Basic Social infrastructure (health, police, kindergarten)	100% of residents within 10 min of key services
Community Engagement Index	60% participation in local initiatives

Energy

KPI	Target Value
Electricity Consumption per Household	Below 2,500 kWh per year
% of Buildings Retrofitted	50% of existing buildings by 2030

Housing

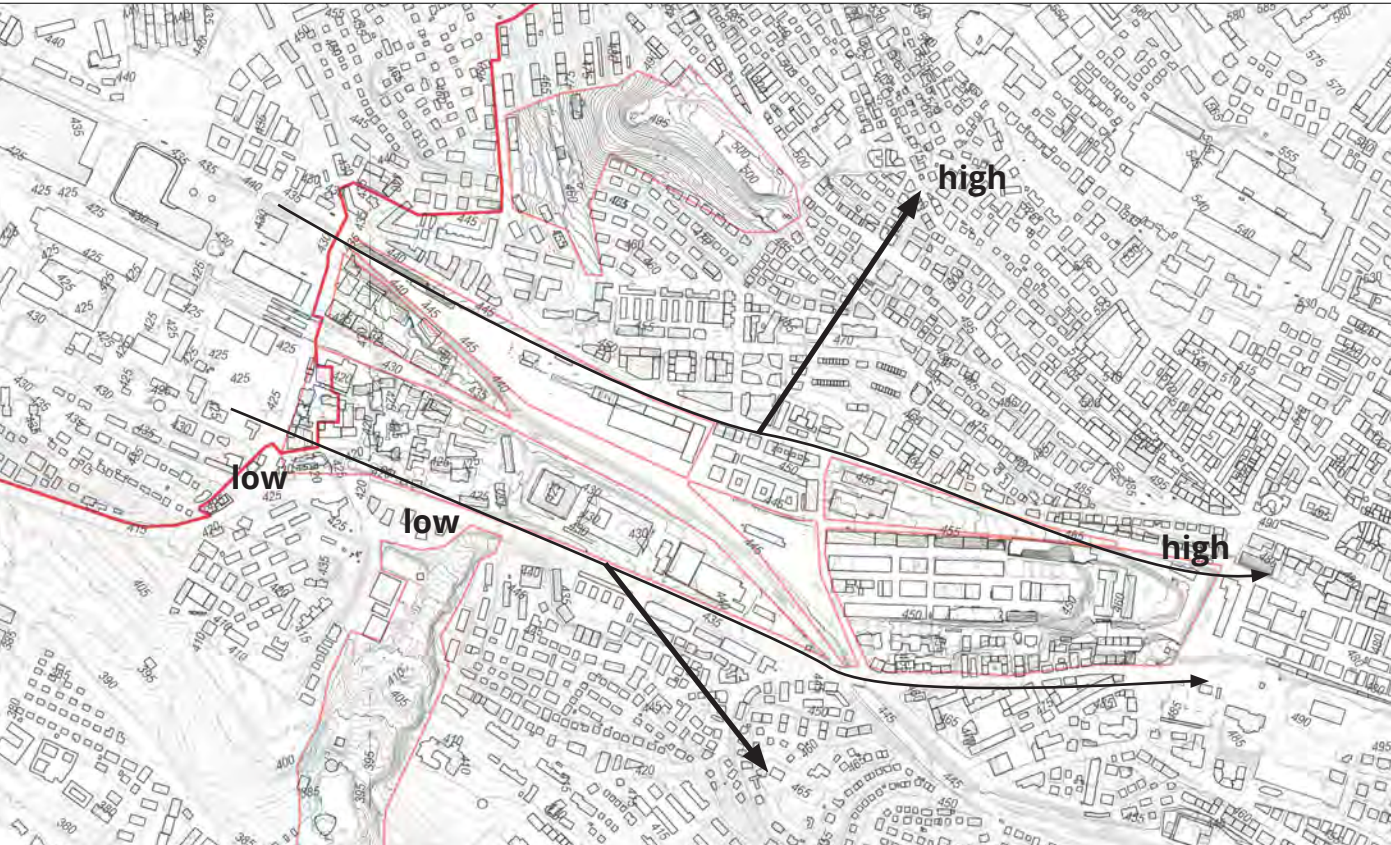
KPI	Target Value
Typology mix	50% multi-family, 30% mixed-use, 20% single-family
Density	Minimum 150 residents per hectare
FAR	2.0
Average Household Size	2.5 people per dwelling
Affordable Housing	10% (usually established by the Municipality)

4 Urban analysis|

4.5 Situational Analysis & Diagnosis

4.5.3 GIS analysis | Topography

Figure 16. Topography of Malley-Sébeillon



Source: Ville de Lausanne, 2024

Geographic features

Q03 covers approximately 104 hectares and is located adjacent to the city center neighborhoods, Q01 and Q02, as well as the commune of Prilly to the west. It is also in close proximity to the Malley *ecoquartier*, as well as the trendy and fashionable districts of Flon. The area is divided by major transport routes: the Geneva-Lausanne highway runs below Sévelin. and the Rue de Genève above.

The topography of Q03 is characteristic of Lausanne, with its hilly terrain. The Martinet quarter sits at a lower elevation, ascending toward Sébeillon. The Valency area, located on a hilltop, offers panoramic views of the historical areas in Quarter Q01 from Sébeillon.

Sébeillon itself lies in a relatively low-lying area and features several stairs as entrance to the area. However, only one has received a design solution appropriate to its urban context.

Figure 17. Transport scheme of Malley-Sébeillon



Source: Ville de Lausanne, 2024

Transport

The area is intersected by multiple railway lines, some of which are no longer operational, including those leading to the Sébeillon Hall.

- Boundaries of study
- buildings
- highway line
- 4
- 5
- 6
- railway-line
- parking

4 Urban analysis|

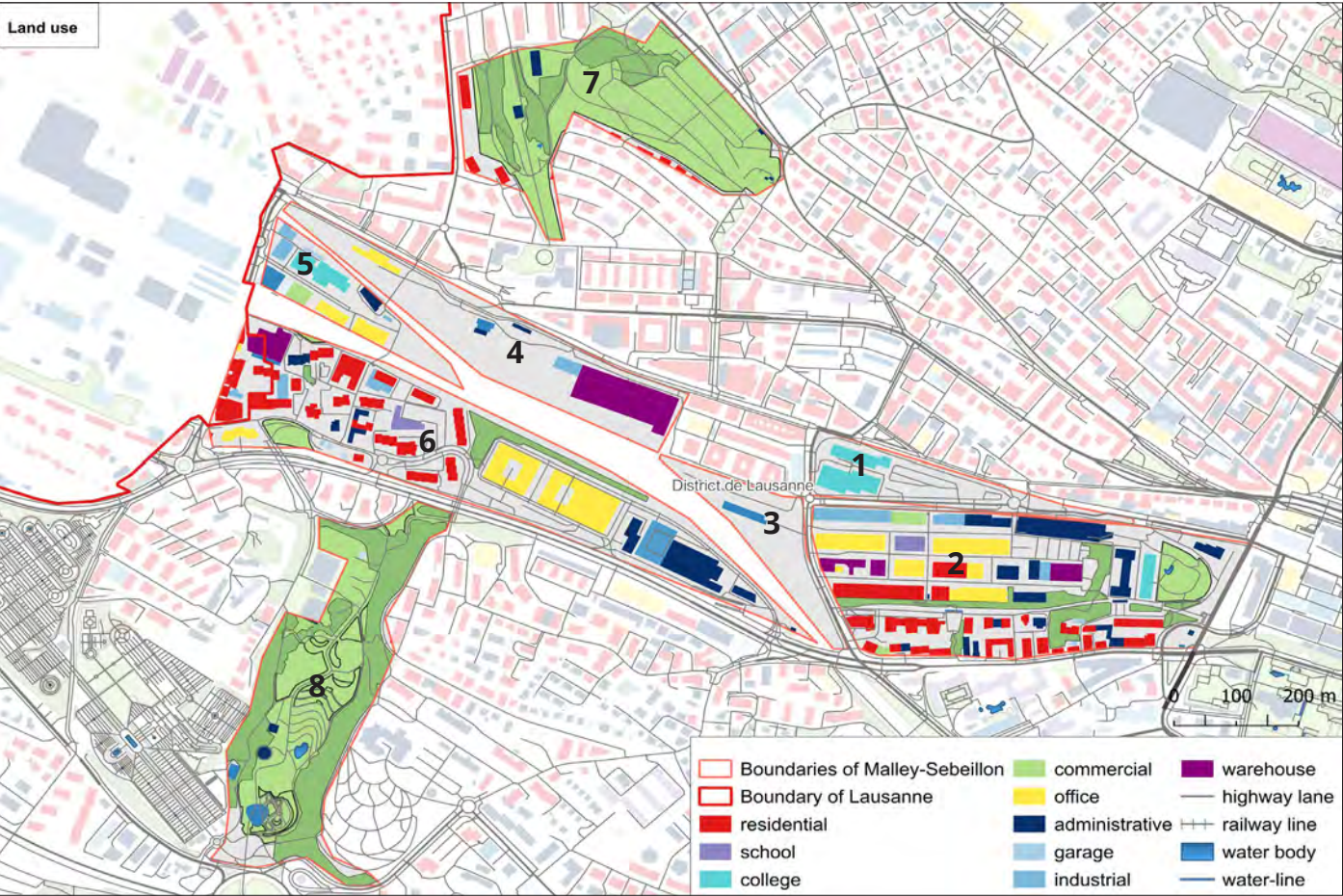
4.5 Situational Analysis & Diagnosis

4.5.4 GIS analysis| Land-use &housing stock

Housing and building stock

The building stock in Western Neighborhoods aligns with the general housing trends of Lausanne, with similar vacancy rates and rental percentages (0,67% and 90% respectfully). The area comprises a total of 183 housing units, with apartments featuring 3 rooms being the most common, accounting for (~35% of the stock). Older buildings constructed before 1960 dominate the housing stock, representing approximately 50%, which indicates a significant need for renovation.

Figure 18. Land use of Malley-Sébeillon.



Source: Adaptation of Open Street Map Data, QGIS, 2024.

KPI. Housing

- 0,5 % of affordable housing (1 building of 4 floors, Sévelin area);
- Density - 104 ha/he;
- Typology mix - 35% Multi-Family, 65% - Single Family;
- Average Household Size - 1 person per family;
- Average FAR - 1.0

Typology of residential houses	Individual	High-rise	Middle-rise
41	3	10	28

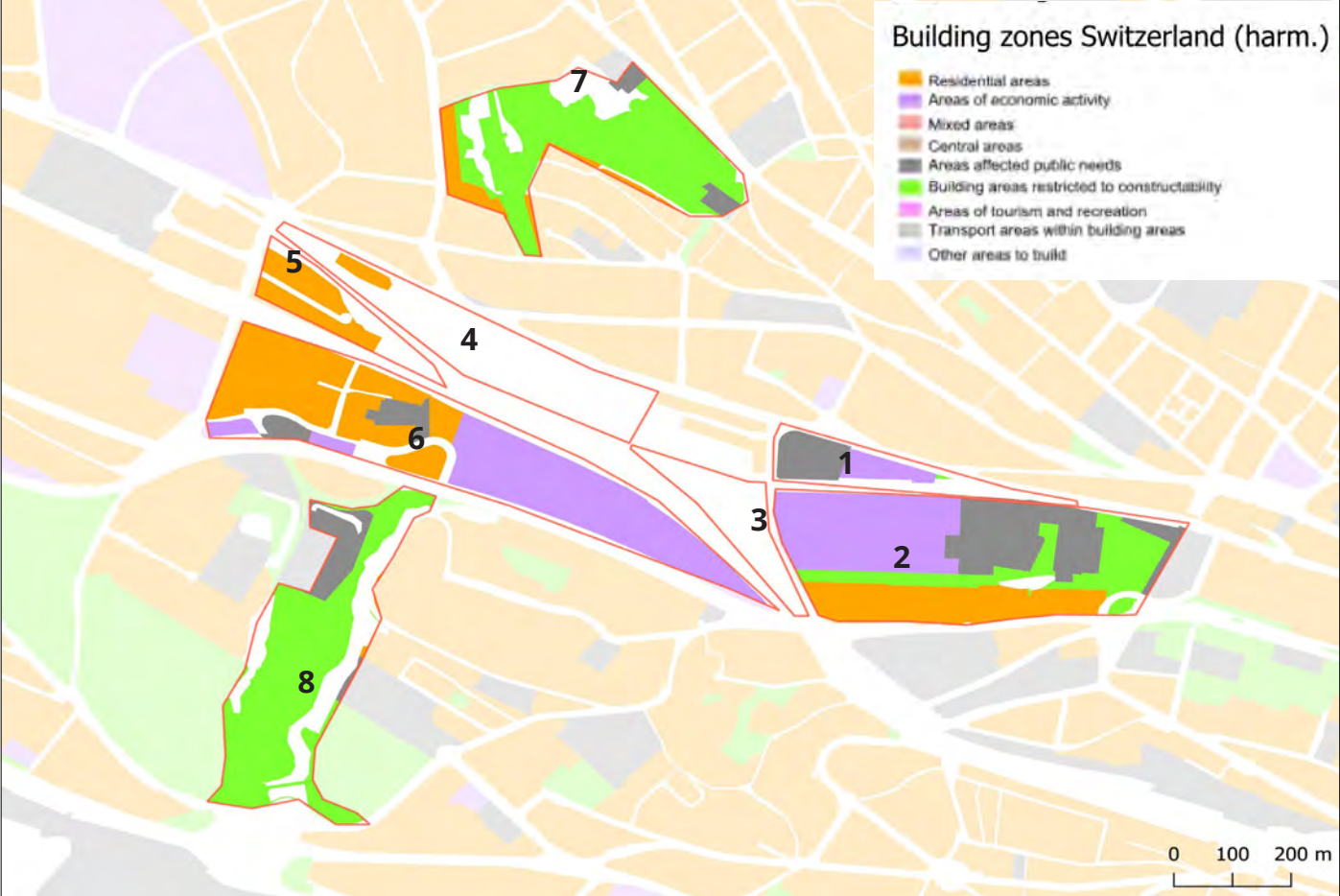
Source: Open Street Map, 2024.

Figure 19. Housing stock of Malley-Sébeillon.

Par-cels	Area	Area of buil-dings	Density of buildings % (Foot-print/ Area)	Level	Nu-mer of buil-dings	Buildings in Land use	FAR (according to average height* of buildings in each designated areas) Volume/Area
1	2,9 he	0, 5 he	17,7%	low	2	College – 100% (administrtraive)	0,7 (low)
2	14,5 he	5,2 he	36,0%	medium	83	Industrial&Logistics – 8.2%; Commercial – 1.58%; Admi-nistrative -14.2%; Multistorey residential – 17%; Service, Culture – 58%	1.8 (good)
3	2,5 he	0,1 he	4,1%	low	10	Commercial – 100%	0,04 (low)
4	6,1 he	1, 6 he	20,5%	medium	10	Sébellion – Industrial – 100%	0, 2 (low)
5	2,6 he	9, 9 he	37,8%	medium	12	Administrative – 52%; Indus-trial – 16%; Commercial – 20%	1.9 (good)
6	1, 4 he	4,4 he	31,4%	medium	47	Residential -34,01%; Indus-trial - 4,10%; Administrative - 30,05%	1.56 (medium)
7	9,8 he	0,3 he	3,6%	low	12	Residential – 67%; administra-tive - 33%	
8	10,3 he	0, 2 he	2,2%	low	6	Administrative – 100%	

Source: Adaptation of Open Street Map Data, QGIS, 2024.

Figure 20. Building zones** Switzerland, Federal Office for Spatial Development (ARE).



Source: Adaptation from the Official Federal Government Site Services <https://map.geo.admin.ch/>, QGIS, 2024.

* Data obtained from field studies

** The «Building Zones Switzerland (harmonized)» standartized dataset is an official geospatial dataset provided by the Federal Of-fice for Spatial Development (ARE) in Switzerland. This dataset compiles geodata on building zones from cantonal offices responsible for spatial planning, as of January 1, 2022. The harmonization process involves standardizing the diverse cantonal zoning types into nine primary usage categories, in accordance with the minimum geodata model on land use planning.

4 Urban analysis |

4.5 Situational Analysis & Diagnosis

4.5.5 GIS analysis | Social Infrastructure and Public Spaces

Social Infrastructure

The infrastructure for basic services in the area is highly developed, although it is sometimes distributed unevenly across different neighborhoods, more in Sévellin and less to Avenue du Chablais. Still several key streets, such as Avenue Jules Godin and Avenue du Chablais, act as hubs for a variety of services, including pharmacies, supermarkets, repair shops, cafés, and more. Additionally, several large shopping centers are located within a 10–20 minute walking distance from the western neighborhoods. Therefore, it can be concluded that the area’s commercial infrastructure is more than adequate. As for the availability of kindergartens, schools, and sports centers within Q03, basic information is accessible through OpenStreetMap services (see Fig. NN). However, more accurate and detailed data should be obtained from the local administration. It is worth noting that there is a lack of infrastructure for children, such as kindergartens and playgrounds, within the immediate area. Nonetheless, this gap is partially offset by the presence of approximately six kindergartens and two playgrounds within walking distance or near public transport stops.

Figure 21. Social Infrastructure within Malley-Sébeillon Area.

Kindergartens	Playground	Sport activities	Pharmacy/ Clinics	Police
Inside				
0	2	12	1/0	0
In proximity 100 m				
6 (200-2400 m)	10	33	11/3 (average 600 m)	2 (600-1200 m)

Source: Adaptation of Open Street Map Data, 2024.

KPI. Social

- Recreational Facilities - 1 play-ground within area; 2 - within 10 min walk (<800 m)
- Access to Basic Social Infrastructure - 100%
- Community Engagement Index - 1.26%

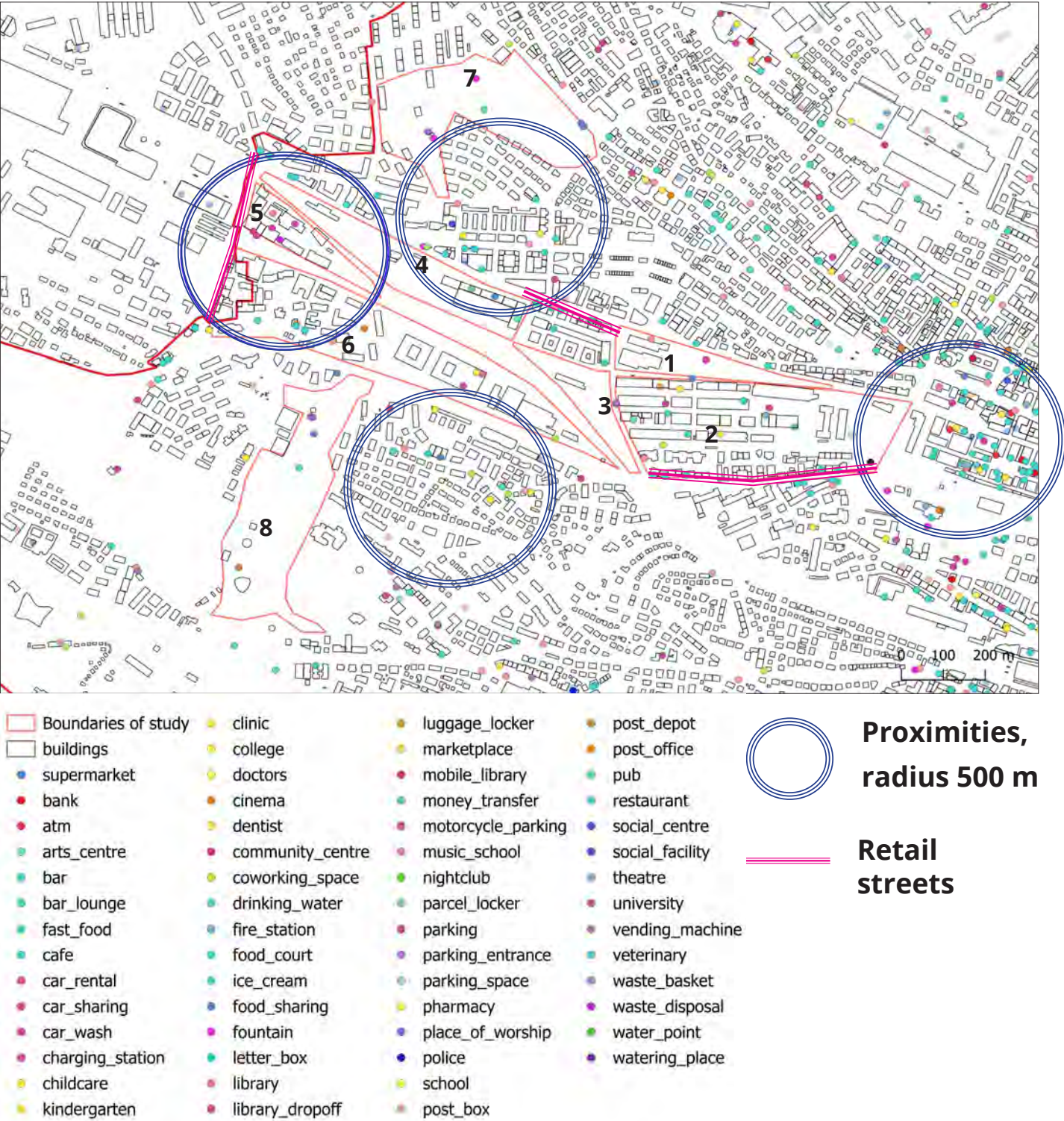
KPI. Economy

- % of Non-Residential Activities - 24%
- Number of Jobs per Resident - 1.06



Street of Sévillan
Credits: authors

Figure 22. Social Infrastructure within Malley-Sébeillon Area.



Source: Adaptation of Open Street Map Data, QGIS, 2024.

4 Urban analysis |

4.5 Situational Analysis & Diagnosis

4.5.5 GIS analysis | Social Infrastructure and Public Spaces

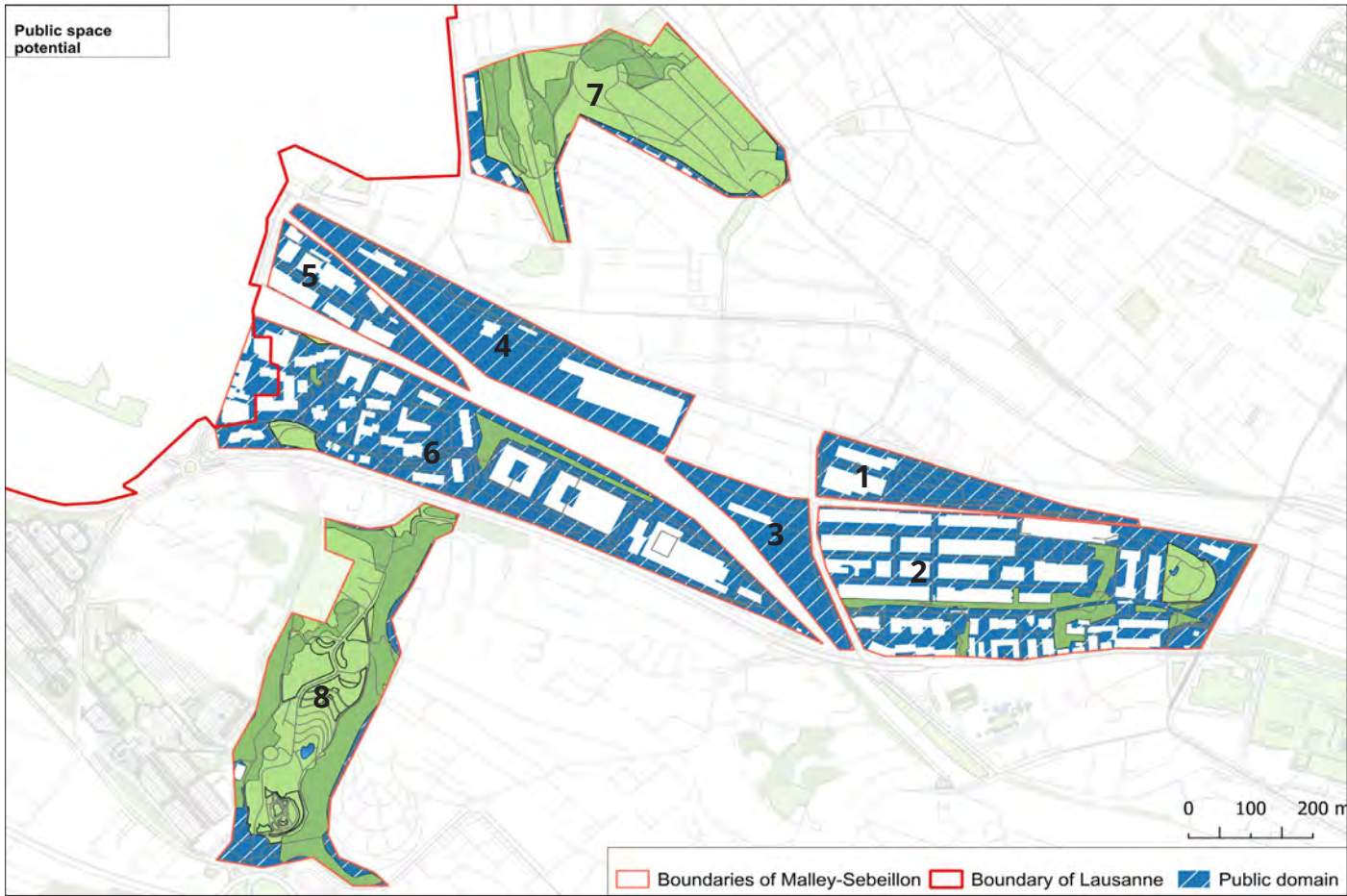
Public spaces

The public spaces in Malley-Sébeillon are characterized by their potential to become highly dynamic, yet they remain underdeveloped. These spaces are fragmented and un-derutilized, with much of their potential untapped and limited to formal design considerations. The area encompasses around 29 hectares (62% of the total area) of public domain —essentially voids between buildings - that could potentially be transformed into public spaces.

Thus, large open parking lots present an opportunity to be converted into parks or squares. Similarly, semi-urban green zones lack cohesion and strategic integration into the broader urban fabric.

Notably, the concentration of benches in the Sébeillon warehouse zone illustrates a dynamic social intersection and active human presence, indicating organic use despite the industrial context. These characteristics underscore the significant opportunity to enhance the development of common public realm, transforming the neighborhoods into a vibrant and engaging urban environment.

Figure 23. Public Domain of Malley-Sébeillon,



Source: Adaptation of Open Street Map Data, QGIS, 2024.

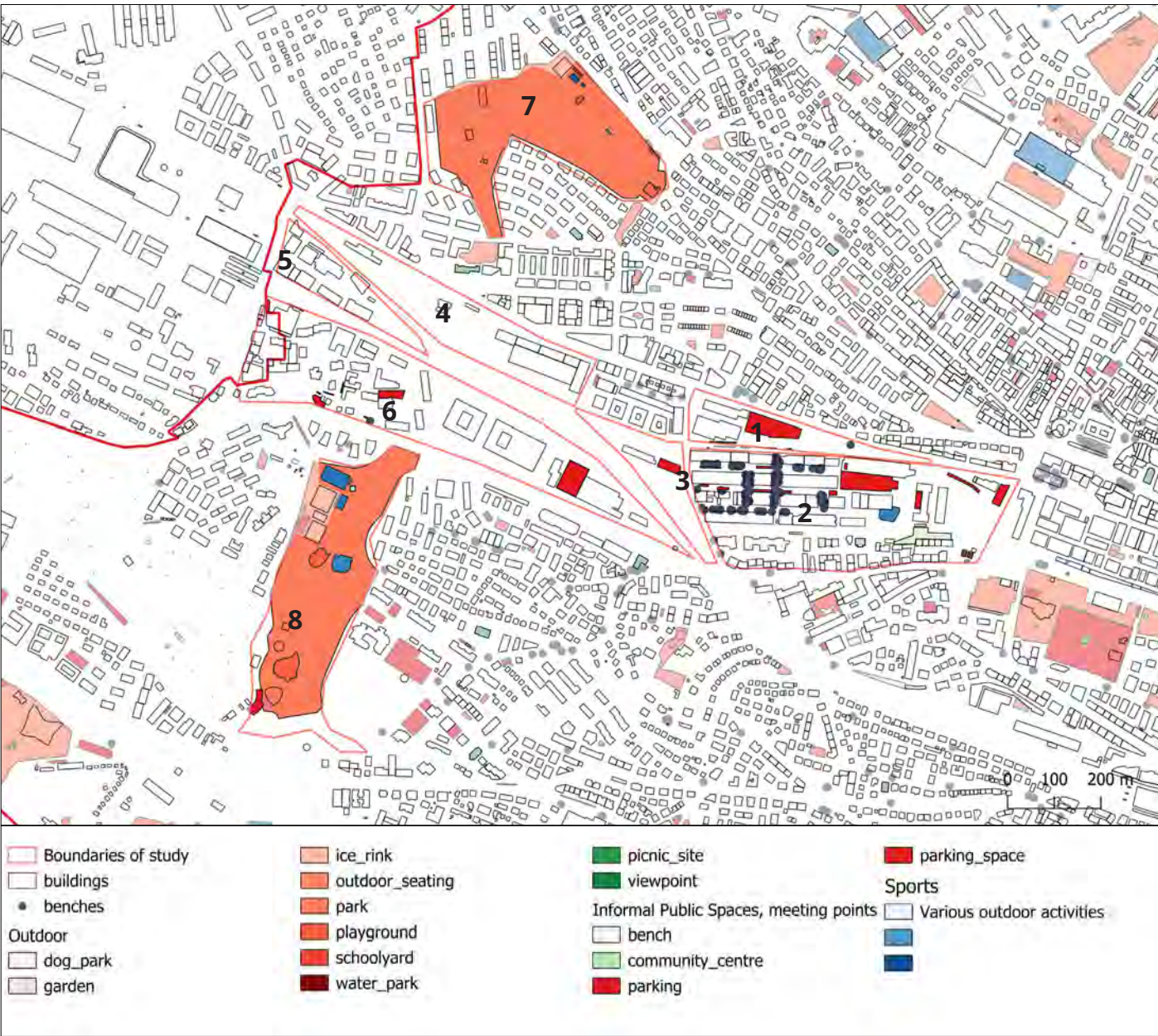
KPI, Social
Public Space Ratio - 43 %
with parks of Valency; 24,6%
without parks



Community gardens in the upper side of Sévèllin

Credits: authors

Figure 24. Public Spaces Infrastructure.



Source: Adaptation of Open Street Map Data, QGIS, 2024.

4 Urban analysis |

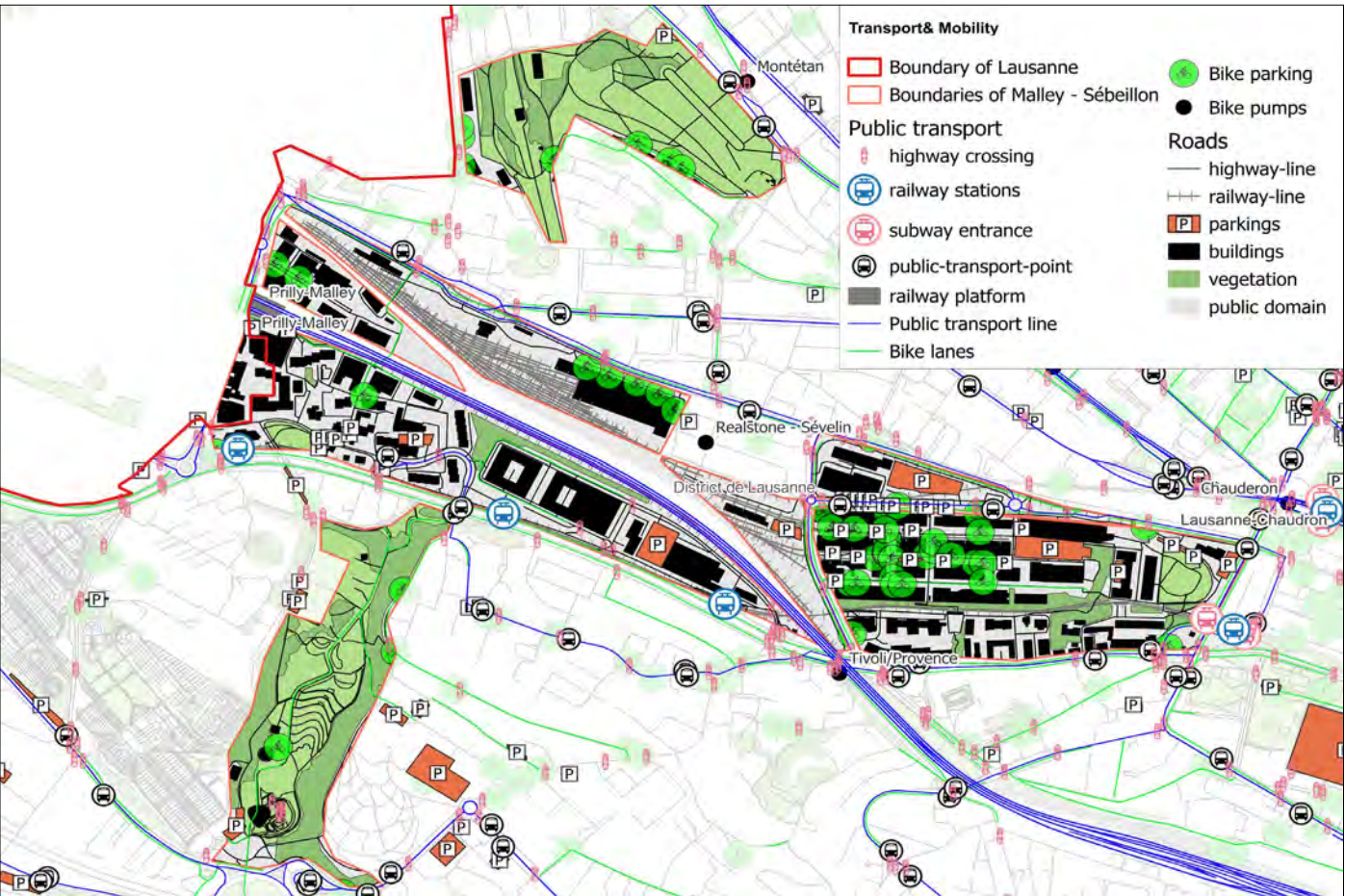
4.5 Situational Analysis & Diagnosis

4.5.6 GIS analysis | Transport&Mobility

The area is divided by major transport routes: the Geneva-Lausanne highway running below and the Rue de Genève above. However, the district also enjoys excellent public transportation, including the Prilly-Malley and Lausanne CFF stations, metro line M1, and the upcoming tram line, all of which enhance sustainable mobility for residents and workers.

It is possible to conclude that prevalence of economic activities over residential make the Q03 a focal point for commuting home - work routine. The visual data about parking allocations provided by *Ville de Lausanne* (2024) confirms this hypothese. The bicycle infrastructure is quite developed for this area, however it lacks multidimensional connectivity, i.e. in perpendicular direction from south to north and on the contrary. The information regarding noise from road and rain traffic (day/nighttime) informs that around 3% of residential areas is exposed to noise, incl. in the night-time to loads greater then 50dB.

Figure 25. Public Transport scheme of Malley-Sébeillon.



Source: Adaptation of Open Street Map Data, QGIS, 2024.

KPI. Transport

Public Transport Network Coverage - 100%

Public Transport Network Access - 100 %

Pedestrian Infrastructure - 100%

Walkability* - 40/100

The walkability methodology is based on J. Speck's theory of walkability (2012), which encompasses four key criteria for pedestrian networks: Useful, Safe, Comfortable, and Interesting. The baseline evaluation was conducted through a field study.

Figure 26 Zone of moderate traffic



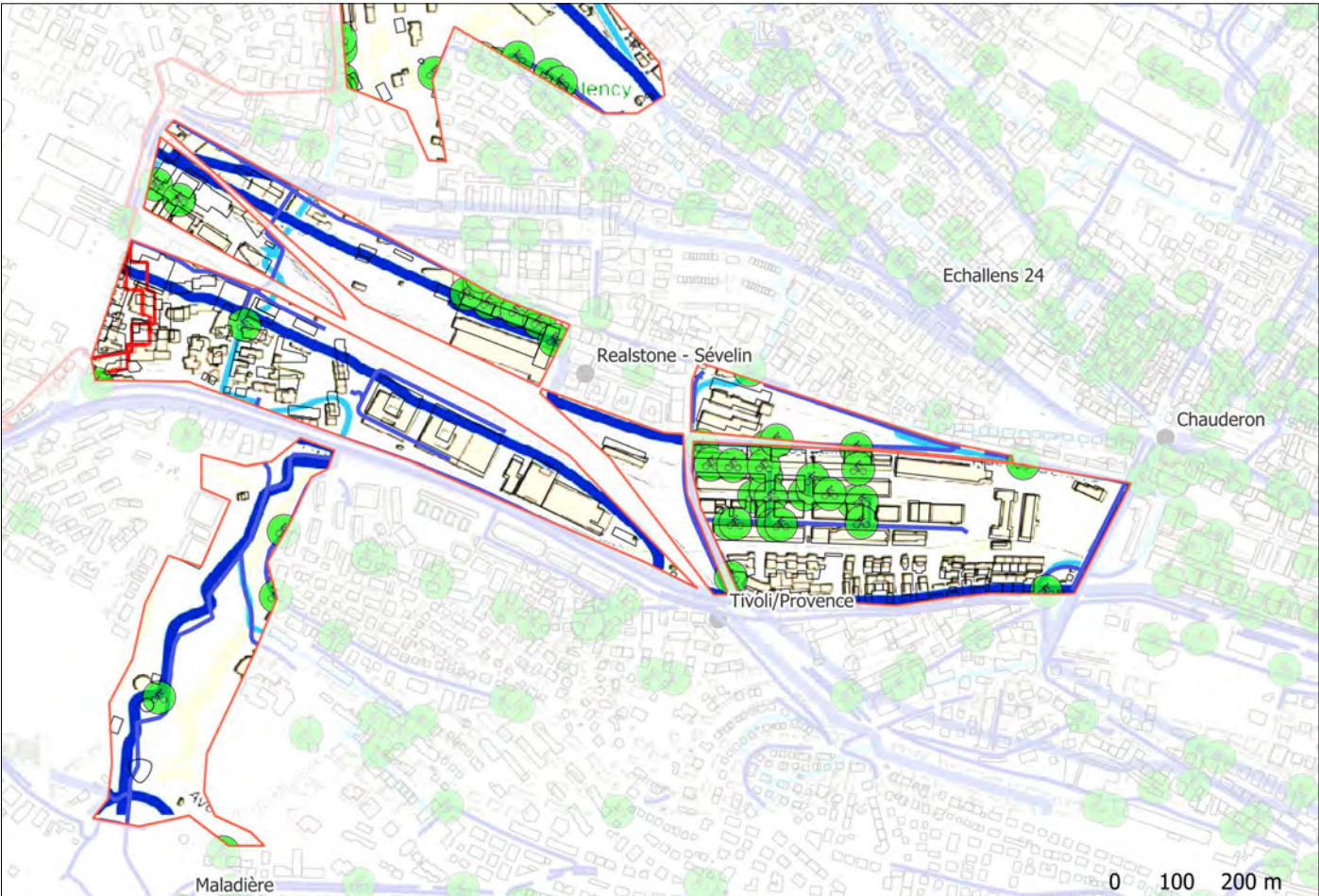
Source: Adaptation of Open Street Map Data, QGIS, 2024.

Figure 27. Infrastructure of Malley-Sébeillon.

Bus stops	Metro Stops	Train stations
Inside		
9	1	4
In proximity 300 m		
15	3	1

Source: Adaptation of Open Street Map Data, 2024.

Figure 28. Bicycle infrastructure of Malley-Sébeillon.



Source: Adaptation of Open Street Map Data, QGIS, 2024.

- Boundaries of study
- buildings
- Bicycle lanes
- Bicycle Service
- Bike Parking
- Bike Pump

4 Urban analysis |

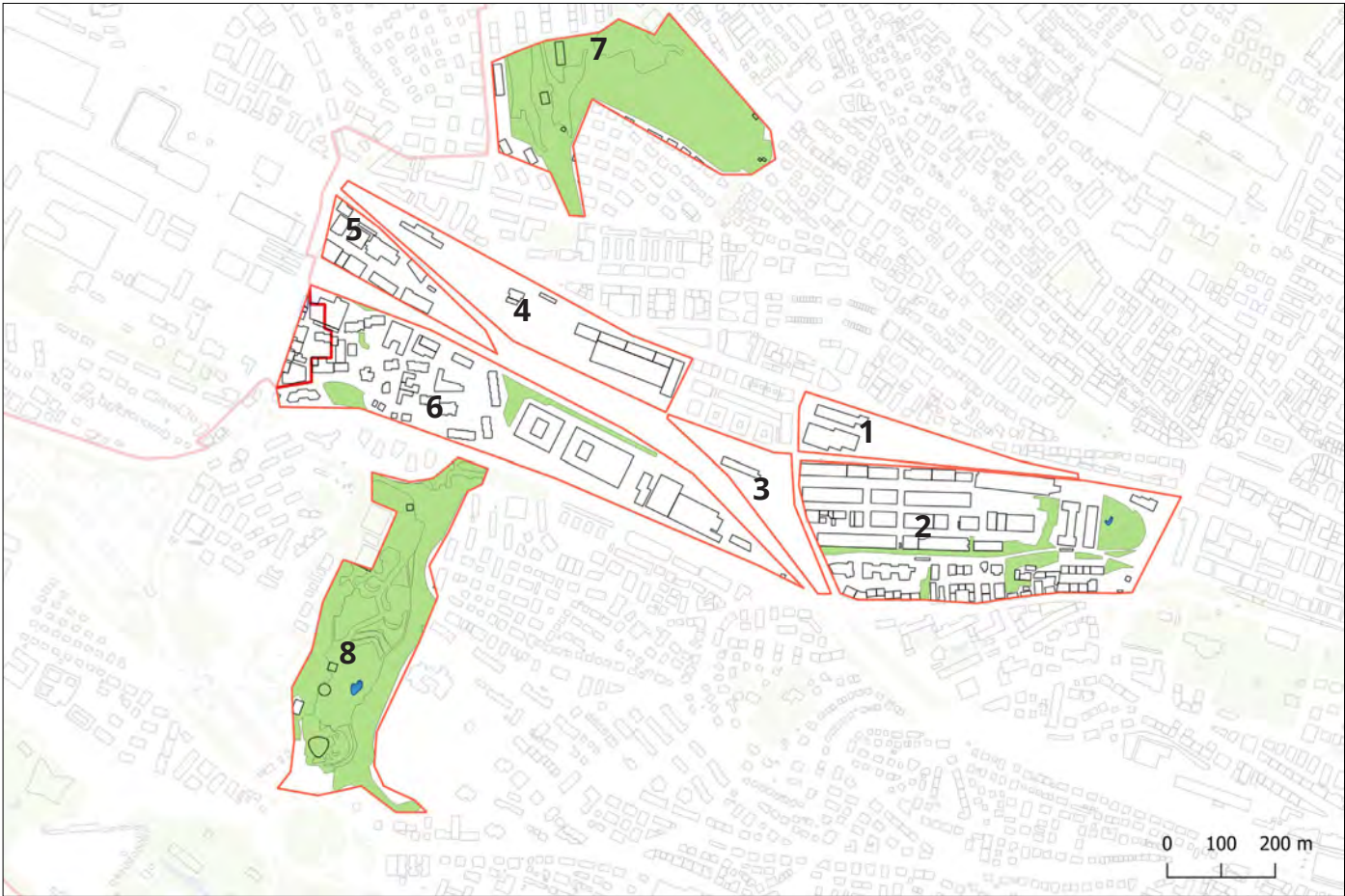
4.5 Situational Analysis & Diagnosis

4.5.6 GIS analysis | Vegetation

Vegetation

Green areas account for approximately 8,47 he (19%) of the total area, excluding the parks of Valency. Data provided by local government indicates that the distribution of vegetation across the neighborhoods is uneven. Large portions of the area are devoid of greenery, and many open public spaces are used solely as parking lots. This lack of vegetation, combined with the extensive hardscapes, suggests that the area is significantly impacted by urban heat island effects (see Urban Heat Map, *Ville de Lausanne*, 2024).

Figure 29. Vegetation of Malley-Sebeillon.



Source: Adaptation of Open Street Map Data, QGIS, 2024.

- Boundaries of study

buildings
- Vegetation

Figure 30.

	Area	Urban Tree Ca- nopy	Green area	Impervious surfaces
1	2,9 he	10%	0%	100%
2	14,5 he	25%	13.4%	80.76%
3	2,5 he	0%	0%	100%
4	6,1 he	0%	0%	100%
5	2,6 he	2%	0%	100%
6	1,4 he	25%	5.84%	94.16%
7	9,8 he	85%	95%	5%
8	10,3 he	90%	98%	2%

Source: Adaptation of Open Street Map Data, 2024.

KPI Environnement

Noise Exposure -
98%<50dB

Green Areas -7.17m2/hab

Green Area Accessibility -
50%

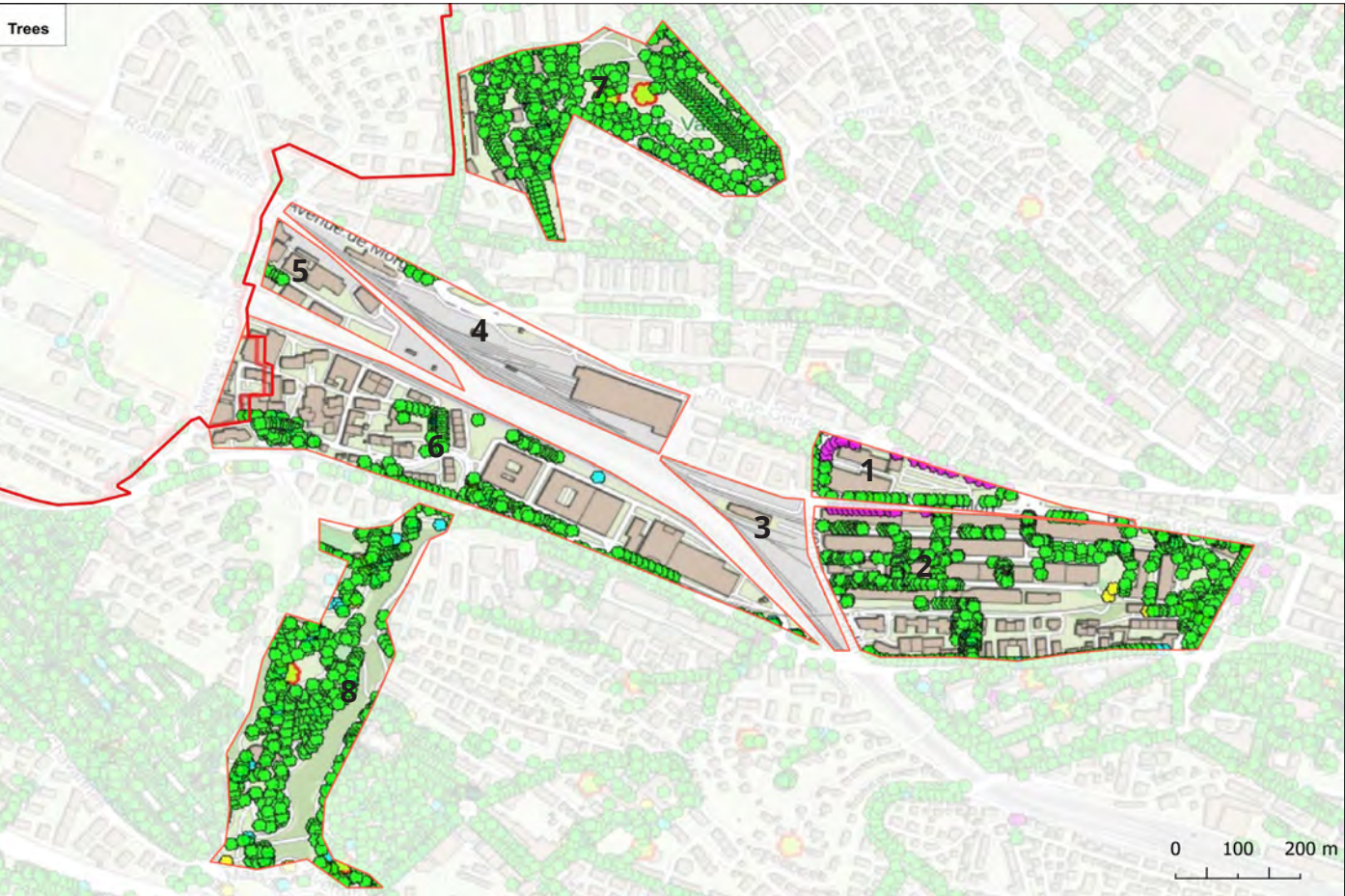
Pervious surfaces - 19%

Urban Tree Canopy Cover
- average 10.3%

Figure 16 : Imagerie satellite USGS/NASAN Landsat 8 (NDVI) – bandes 4 et 5, température du sol – bande 11)

Lausanne Heat Map, (Source: *Ville de Lau-
sanne – Modelisation Climatique*, 2024)

Figure 31. Urban Tree Canopy.



Source: *Ville de Lausanne*, guichet cartographique, 2024.

4 Urban analysis |

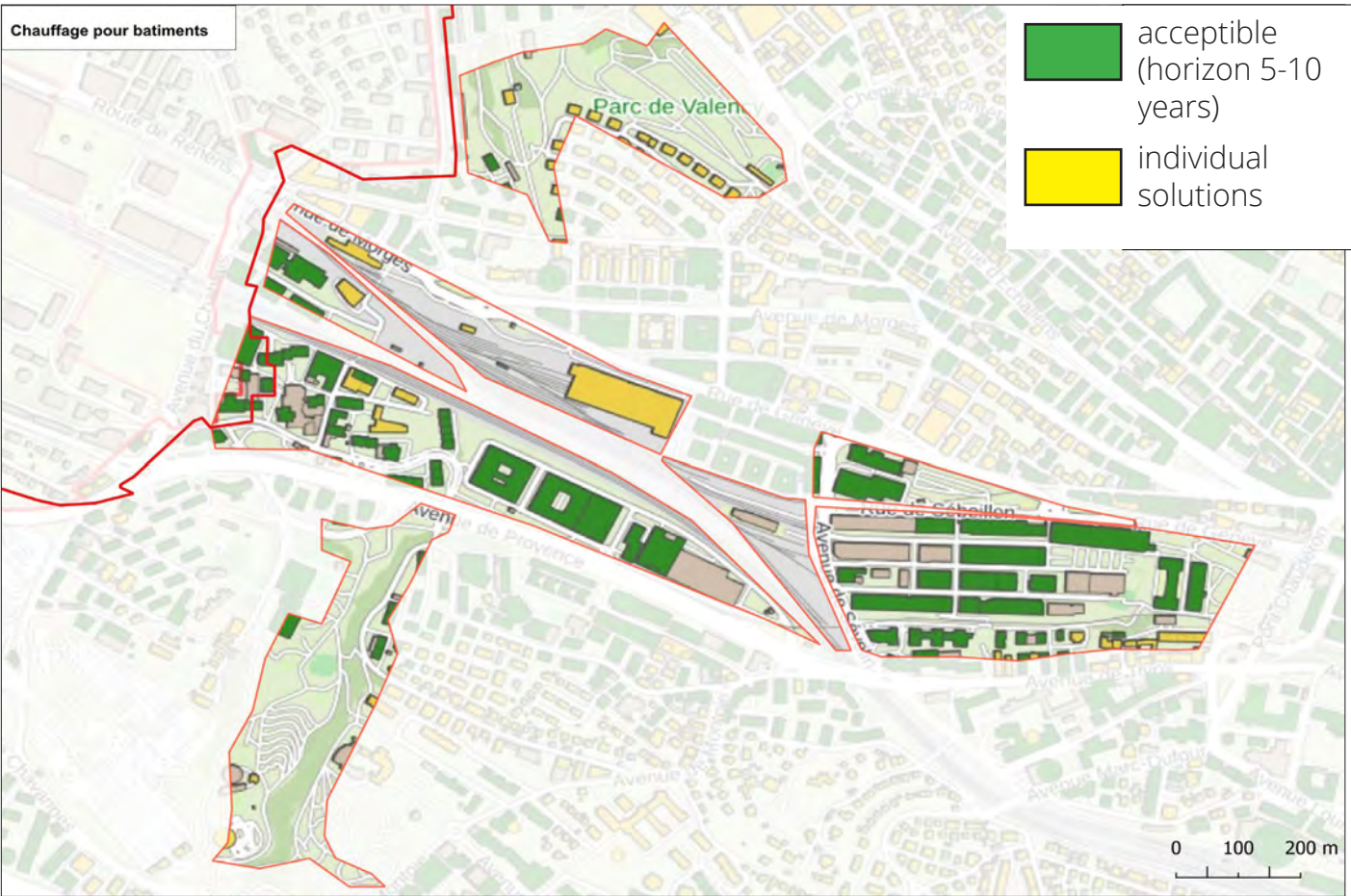
4.5 Situational Analysis & Diagnosis

4.5.7 GIS analysis | Energy profile

Heating

As mentioned, Lausanne is pursuing ambitious goals to become a city of energy excellence. One of its top priorities is to achieve a complete transition to renewable energy sources. As Lausanne actively advances its energy transition efforts, Malley-Sébeillon is set to benefit from the city's initiatives.

Figure 32. Evaluation by SIL (*Services Industrielles de Lausanne*) of potential for ditrict heating.



Source: Ville de Lausanne, 2024

Figure 33. Profil of annual energy consumption in Malley-Sébeillon (in MWh)*

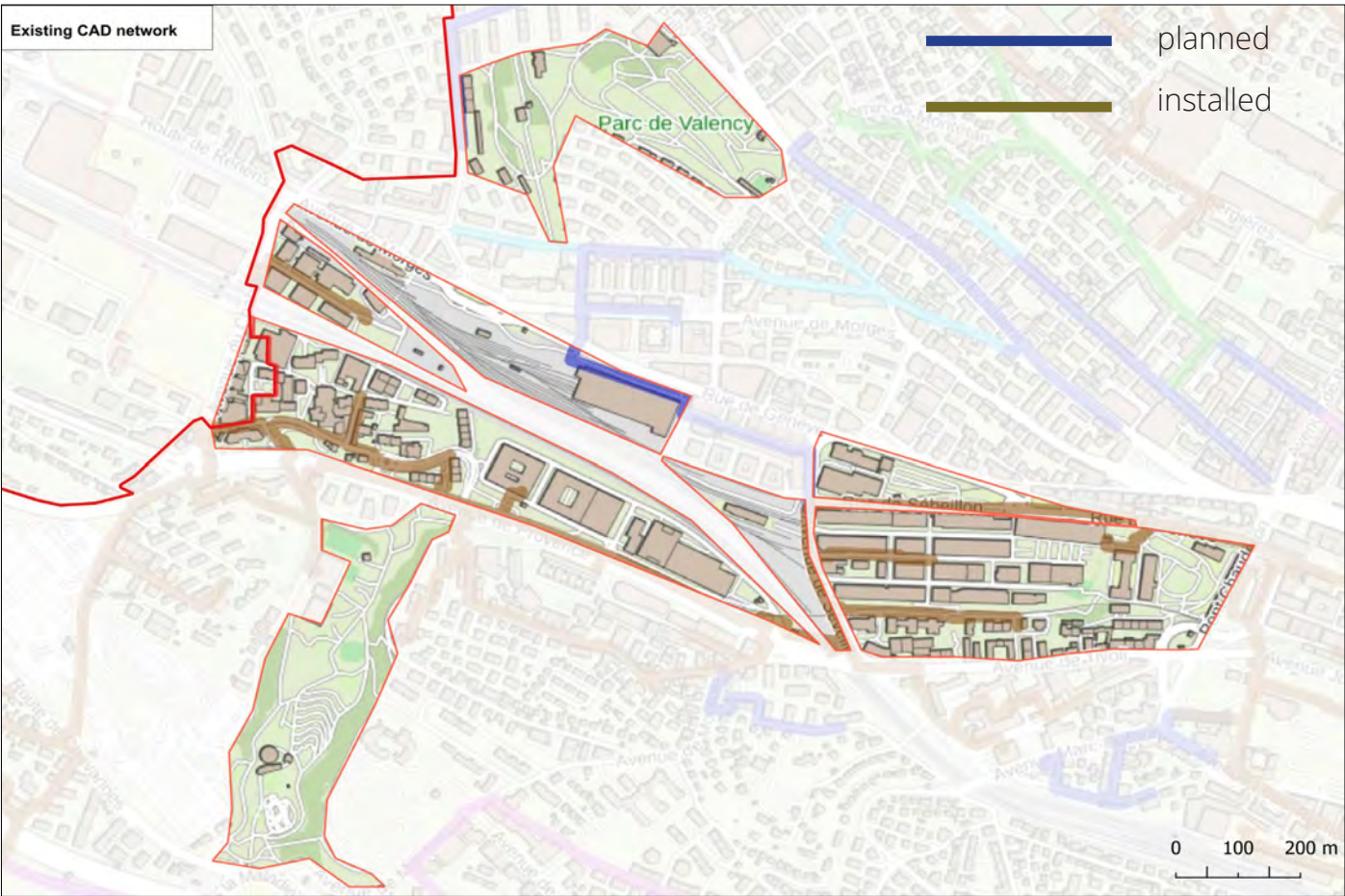
Households	Industry	Services	Transport	Total of groups	Selling to communes	Total
24 544	9 039	25 341	4 779	63 814	36 704	100 518

Source: Ville de Lausanne, Bueau statistique de Lausanne, 2024

Total number of households is 6231 (25 544 MWh)

Currently, district heating in Lausanne is powered by 60% renewable energy sources, primarily from the TRIDEL waste-to-energy plant and STEP (wastewater treatment plants). The goal is to transition to 100% renewable energy by 2035. At present, the district heating network extends to Malley, covering a total of 116.1 kilometers, and continues to expand westward.

Figure 34. District Heating Network.



Source: Ville de Lausanne, 2024

KPI Environment

- Per Household Energy Consumption - app. 4099 kWh
- % of Retrofitted Buildings - N/A

4 Urban analysis |

4.5 Situational Analysis & Diagnosis

4.5.8 GIS analysis | Solar Potential

Solar Potential

In 2023, Si-REN, a Municipal-based Renewable Energy Company produced approximately 1 GWh of solar energy. According to its report, several projects have already been implemented in Malley-Sébeillon. This is largely due to the presence of numerous warehouses, which offer a significant amount of surface area—often exceeding 100 m²—suitable for the installation of solar panels. The most part of the neighborhoods roofs is characterised, according to Si-REN as «very good» for installation of solar panels. There is also one Minergie certified building in the area of Malley-Sé-

Figure 35. Evaluation of Solar Potential for Roofs (Low, Medium, Good, Very Good).



Source: Federal Office for the Environment FOEN, 2024

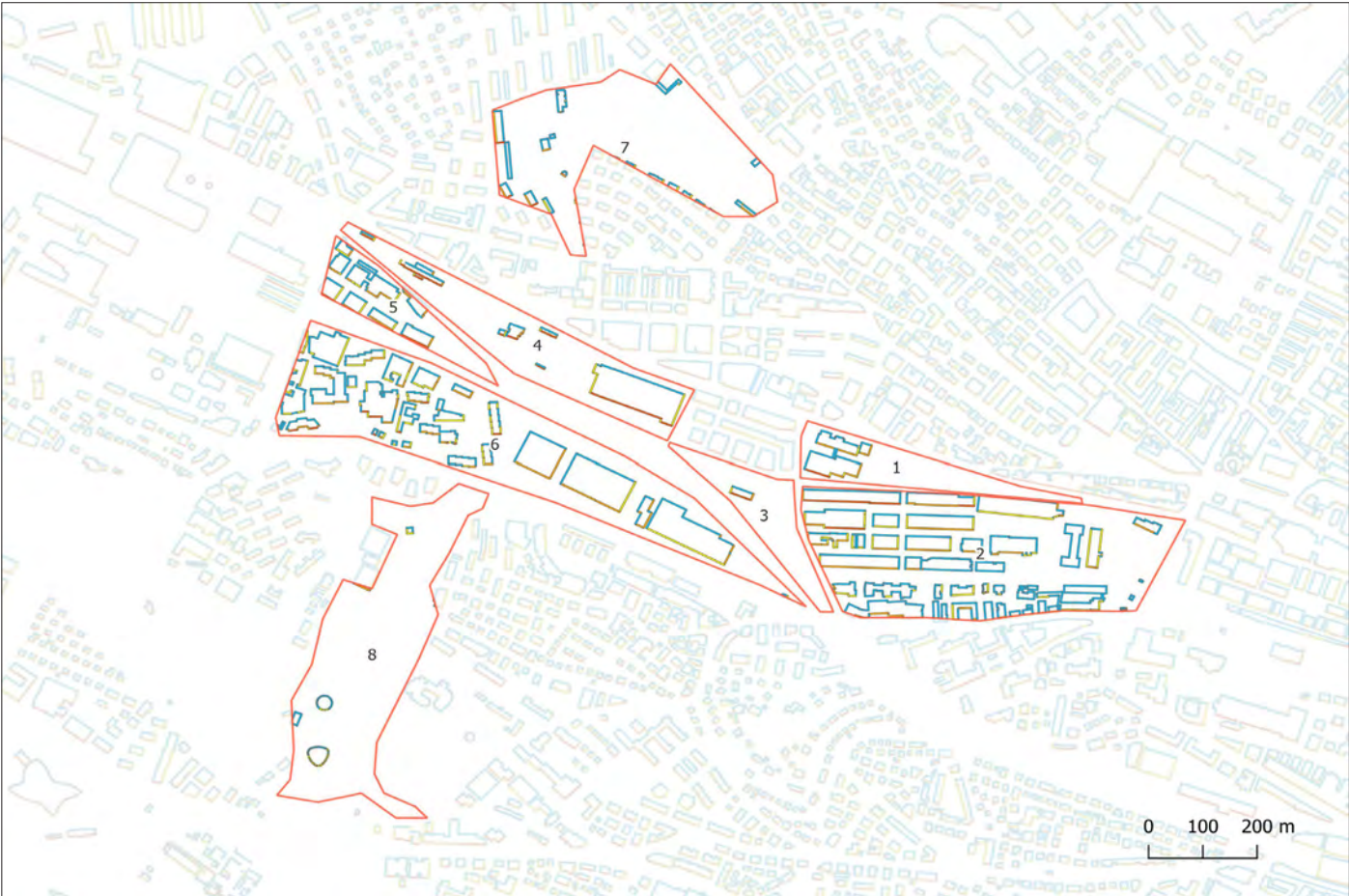
Solar energy: suitability of roofs

- Not defined
- Low
- Mean
- Good
- Very good
- Top

M - MINERGIE certified building

beillon. It is worth mentioning that proximity of Malley neighborhood with its established 2000Wh Society serves exemplary for surrounding developments.

Figure 36. Evaluation of Solar Potential for Roofs (Low, Medium, Good, Very Good).



Source: Federal Office for the Environment FOEN, 2024

Solar energy: suitability of facades

- Not defined
- Low
- Mean
- Good
- Very good
- Top

4 Urban analysis |

4.5 Situational Analysis & Diagnosis

4.5.8 GIS analysis | Geothermal Potential

Geothermal Potential

According to information from Geoservice of Canton Vaud, Malley-Sébeillon has significant potential for geothermal energy. Potential production, accounting for approximately 1 GWh annually, covers around 23% of the energy needs and is located in the Sévelin area.

Figure 37. Geothermal Potential of Mally-Sébeillon.



Source: Geocat.ch, Canton Vaud, 2024

Potential production from geothermal energy

- < 0.5 GWh/an
- 0.5 à 1 GWh/an
- 1 à 5 GWh/an
- 5 à 10 GWh/an
- > 10 GWh/an

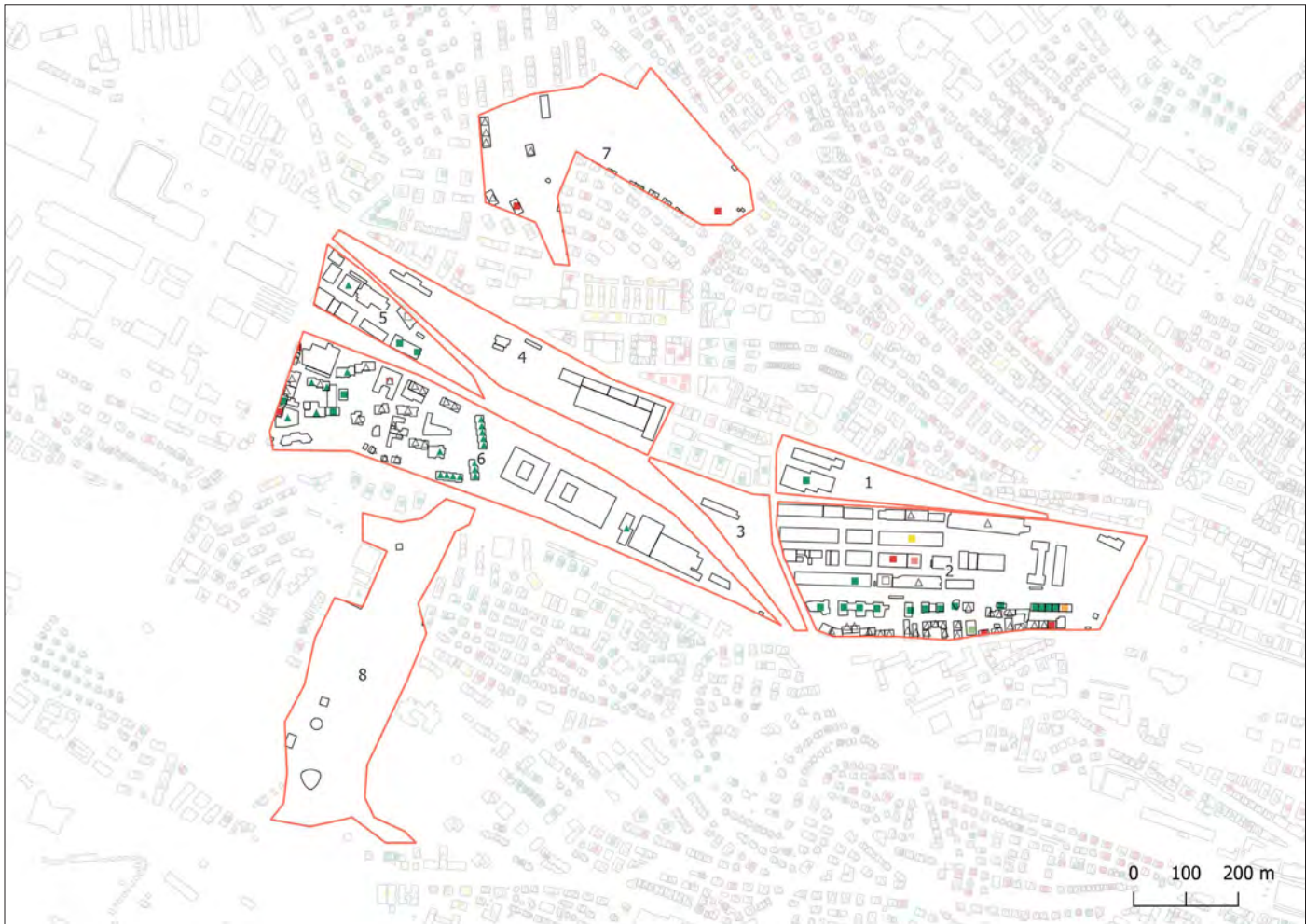
4 Urban analysis |

4.5 Situational Analysis & Diagnosis

4.5.8 GIS analysis | CO2 emissions

According to the Federal Office for the Environment, data on the CO2 building footprint is available for the whole of Switzerland. In the Sébeillon area, a few buildings have very high CO2 emissions, but the majority fall within the low range (5 to 10 kg/m²). This provides a strong foundation for the possibility of transforming the neighborhood into a nearly zero-emission area.

Figure 38. Evaluation of Solar Potential for Roofs (Low, Medium, Good, Very Good).



Source: Federal Office for the Environment FOEN, 2024

KPI Environment

- Per Household Energy Consumption - app. 4099 kWh
- % of Retrofitted Buildings - N/A
- CO2 per building - average 5 to 10kg/m2

Boundaries of study
buildings
CO2 emissions buildings (SIA 380/1)

- No information
- CO2 emissions: 0 kg/m² heated area / year
- CO2 emissions: 0 to 5 kg/m² heated area / year
- CO2 emissions: 5 to 10 kg/m² heated area / year
- CO2 emissions: 10 to 15 kg/m² heated area / year
- CO2 emissions: 15 to 20 kg/m² heated area / year
- CO2 emissions: 20 to 25 kg/m² heated area / year
- CO2 emissions: > 25 kg/m² heated area / year
- CO2 emissions: 0 kg/m² heated area / year; Update date heating system >20 years
- CO2 emissions: 0 kg/m² heated area / year; Update date heating system >20 years
- CO2 emissions: 5 to 10 kg/m² heated area / year; Update date heating system >20 years
- CO2 emissions: 10 to 15 kg/m² heated area / year; Update date heating system >20 years
- CO2 emissions: 15 to 20 kg/m² heated area / year; Update date heating system >20 years
- CO2 emissions: 20 to 25 kg/m² heated area / year; Update date heating system >20 years
- CO2 emissions: > 25 kg/m² heated area / year; Update date heating system >20 years

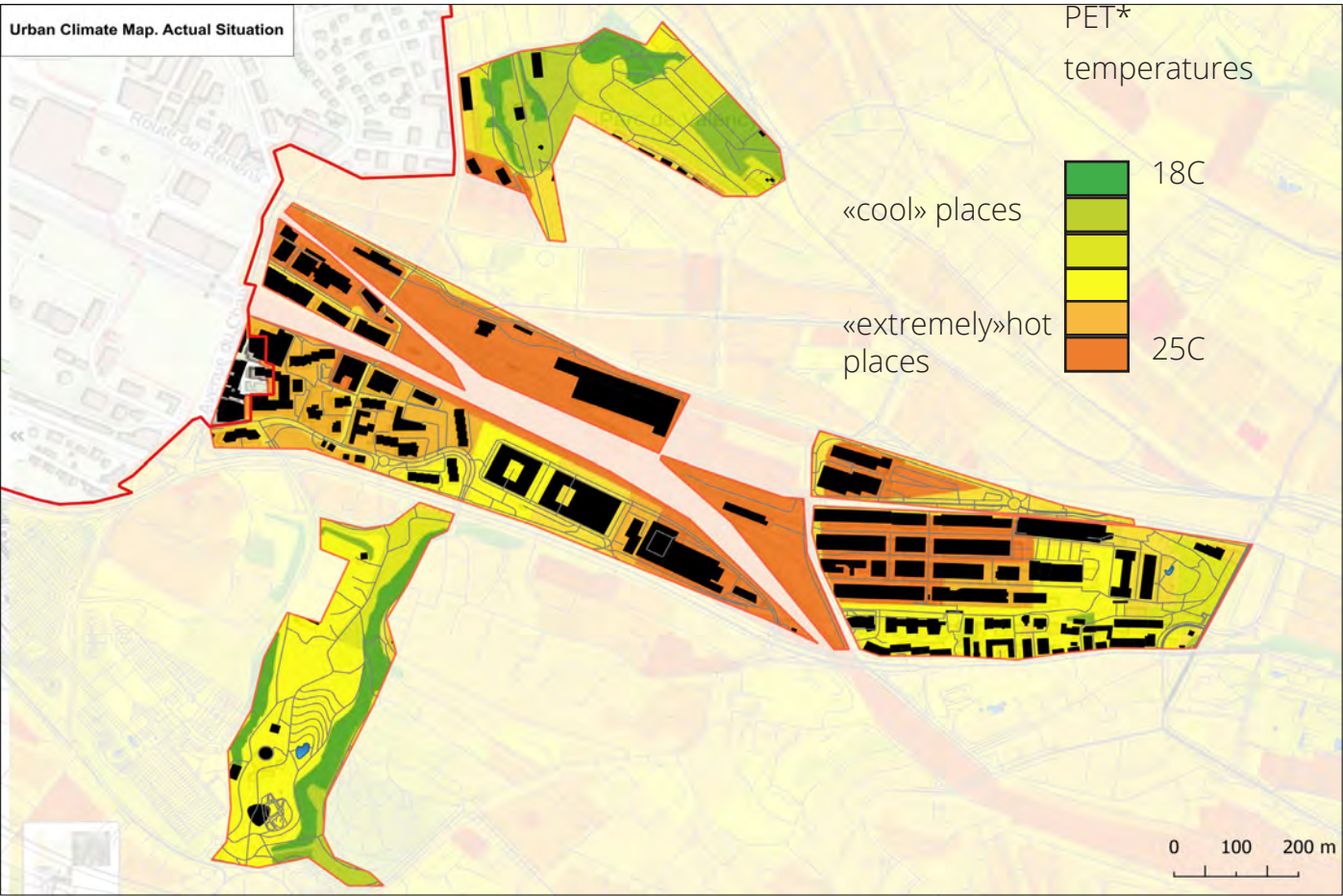
4 Urban analysis |

4.5 Situational Analysis & Diagnosis

4.5.9 GIS analysis | Climate Change Scenario

Decarbonization is a critical concern in Switzerland due to the potentially severe economic and environmental consequences of rising temperatures caused by climate change. As a follow up to IPCC 5 scenarios (IPCC, 2016) in Switzerland it exists two operating scenarios of climate change: 1) rising of temperature between 1.5 and 2 C and in case decarbonization measures would be taken 2) augmentation between 4 and 8 degrees without measures taken. Data from MeteoSuisse (2023) indicates that the country's average temperature increased by 2°C between 1869 and 2010. In case no measures would be taken it will lead to four key impacts identified in the National Program CH 2018 that will have devastating economic and environmental consequences: a) Dry summers b) Reduced snowfall c) Extreme precipitation events d) An

Figure 39. Urban Climate. Actual situation



Source: Guichet cartographique, Ville de Lausanne, GIS, 2024.

KPI. Environment

Pervious - 19%

Impervious surfaces - 81%

increase in tropical days and nights. As rising temperatures pose significant concerns in urban areas and present threats, particularly to vulnerable groups, it was crucial to assess the presence of hot and cool spaces around Malley-Sébeillon.

Figure 40.

	Area	Area of buildings	Green area	Impervious surfaces
1	2,9 he	0, 5	0%	100%
2	14,5 he	5,2	13.4% (19 561)	86.6%
3	2,5 he	0,1	0%	100%
4	6,1 he	1, 6	0%	100%
5	2,6 he	9, 9	0%	100%
6	1,4 he	4,4	5.84%	94.16%
7	9,8 he	0,3	95%	5%
8	10,3 he	0, 2	98%	2%
	Total 49,2 he	22,2 he (45%)		

Figure 41. Urban Climate. Rising for 4C if no measures would be taken | Tropical night/day parameter.



Source: Guichet cartographique, Ville de Lausanne, GIS, 2024.

The Physical Equivalent Temperature (PET) is a widely used indicator in biometeorology to evaluate human thermal comfort in outdoor environments. It quantifies the combined effects of atmospheric and environmental conditions on the human body's thermal state. PET translates these complex interactions into a temperature value that represents how the body would feel under similar conditions indoors.

4 Urban analysis |

Sustainability assessment | Results

According to the criteria for eco-neighborhoods—such as density, functional mix, sustainable mobility, high environmental quality, and intergenerational and social diversity—the indicators for Malley-Sébeillon are currently high in some aspects (access to public transport amenities, commercial infrastructure) and low in others (green, housing, density). In some aspects, they lag behind current standards, particularly in density, sustainable mobility, and intergenerational diversity.

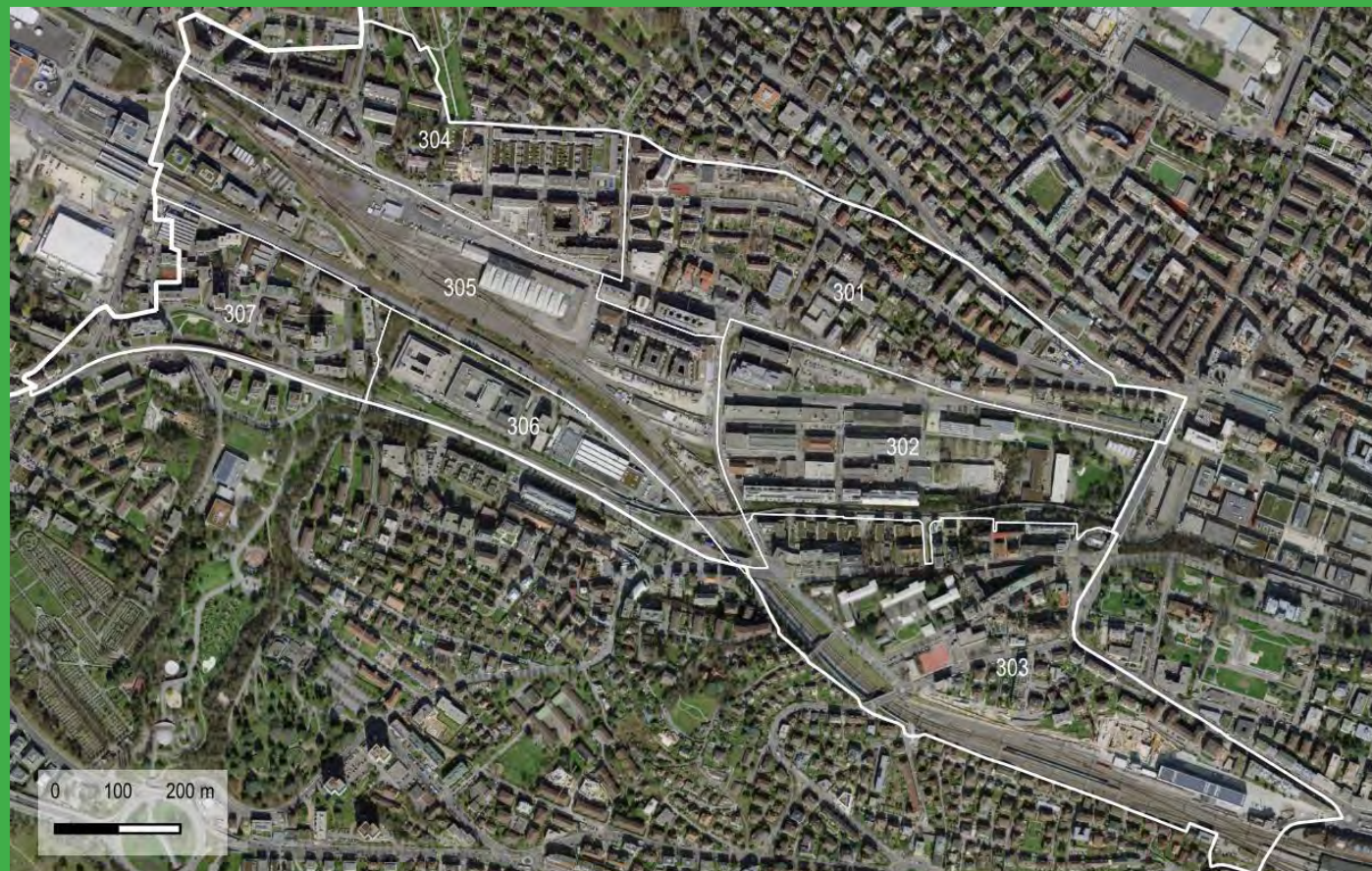
While the population density in the area aligns with the UN's recommended standards for urban density (150 inhabitants per hectare), the built-to-site ratio (FAR) remains very low, around 1,5. This indicates that the land is not being utilized to its full potential, especially considering Lausanne's significant housing shortage. For instance, the current FAR ratios, ranging from 0.04 to 0.7, highlight the urgent need to intensify land use in the area.

An evaluation of the existing social infrastructure reveals that while accessibility is generally satisfactory—averaging a 10-15 minute walk (600–1200 meters)—the quality of these connections remains suboptimal. For example, the walkability of the area is hindered by an uninviting environment, insufficient protections for pedestrians, and the lack of clear, direct pathways. The existing urban layout restricts pedestrian movement, requiring long detours along main streets due to barriers like the railway, which obstructs direct crossings

Although Malley-Sébeillon stands to benefit from Lausanne's energy efficiency policies—supported by several ongoing projects that will ensure the provision of sustainable energy—there remains a notable lack of focus on enhancing quality of life, fostering social inclusiveness, and promoting community-oriented development. Despite social inclusiveness being a key element of Lausanne's urban planning agenda, no substantial proposals have yet been made to address these challenges in the neighborhood.

There is a clear bottom-up demand from the Sevellin's administration for a transformation into a vibrant, dynamic, and high-quality space that accommodates all social groups. However, progress remains slow. A large participatory process is currently underway, and the findings from the initial planning phase have provided the foundation for the current project proposal. These findings, along with the proposed interventions, will be explained in detail in the next chapter.

5 Proposal of Integrated Concept for the Development of a Vibrant and Sustainable Urban Neighborhoods



Credit: Ville de Lausanne, 2024

Concept

Based on KPI calculations, combined with insights from policy documents and field studies, a comprehensive strategy was developed. This strategy serves as a guiding framework for a roadmap to transition the Western neighborhoods, specifically addressing the unique context and needs of the Malley-Sébeillon area.

As outlined by the municipality and administration, one of the central aims for the area is to create a vibrant, mixed-use neighborhood with improved connectivity. In response to this, the core focus of the current proposal is connectivity. In practical terms, this involves developing new pedestrian and bike path systems on new and existing streets.

We believe that streets are essential public spaces that need to be programmed and designed intentionally to foster a connected and vibrant urban environment. While squares and parks are occasional destinations that require specific programming, streets are spaces users experience daily, often unintentionally. The more comfortable and functional the streets, the greater the user satisfaction with their living environment. Streets address everyday needs—shopping, services, repairs, childcare, gyms, or bus stops—and enable users to meet multiple needs efficiently, without long commutes.

The design of the street has the power to shape people's behavior and experiences. Visionary urban design must anticipate behaviors, program public spaces accordingly, and create opportunities for users.

A well-designed street not only serves individuals but also attracts people, generates foot traffic, and increases revenues for local businesses. This creates an economically, socially, and culturally thriving urban space with a distinct and emblematic identity. For this reason, streets form the central typological entity of this proposal. They are co-benefit spaces—when well-designed, street design indirectly, and sometimes directly, supports energy efficiency, social inclusivity, and innovation, aligning with contemporary urban development principles and sustainability goals. Design, in this broader sense, addresses many challenges and can yield both intangible results—such as fostering social connections—and tangible outcomes, like encouraging new businesses to flourish.

The second typological entity we address in this proposal is buildings. In this category, the focus is on implementing energy efficiency measures and promoting mixed-use development—both economically and socially. Additionally, it is critical to densify the urban fabric by introducing new buildings. These new structures should not only meet functional requirements but also create a diverse, emblematic, and aesthetically strong architectural identity. Such an identity will inspire pride and a sense of belonging among residents, making the area a desirable place to live.

The concept of energy efficiency, integrated seamlessly into everyday life, can serve as a unifying element that shapes a cohesive community of residents who prefer this neighborhood for its values, functionality, and vision.

The third element of the current proposal is green space not only in its ecological sense but as a context for economic development. Having seen from the field study grassroot initiatives for communal gardens public space of courtyards of residential areas can become a nice place for local production. However the main idea for this proposal is to enlarge amount of green spaces and make a quarter a real part of “sponge” city so it could lower the load for sewing and stormwater collectors, but bring co-benefits in froms of biodiversity, purified air and mitigation of flood risks.

By addressing streets, green space and buildings holistically in the context of climate change and energy efficiency this proposal aims to foster a vibrant, sustainable and inclusive quarters settled in an area that already has received pioneering inititaitves of malley neighborhood. So Malley Sebeillon will give a continuity to pioneering initiatives of Lausanne in westward direction but at the same time it will keep its social dominance and accessibility to a wider range of cities population.

A framework encompassing key development tracks and actionable measures :

1. Socio-Economic measures

Mixed-Use Composition and Public Space Development

The proposal prioritizes a mixed-use composition that integrates residential, commercial, cultural, and recreational functions to ensure a dynamic and balanced urban environment. Streets are envisioned as democratic public realms, promoting inclusivity and social interaction. This approach is supported by:

Compatibility of Functions: According to Hillier (2004) the street should ensure that its functions (i.e. shops) are «mentally» compatible i.e. the similar or complementary spectrum of consuming, same target audience and others. This is how the streets become active. Active ground floors integrate retail and shared community spaces.

Street-Level Design Principles: The compatibility of streets should be supported by unified design code of streets: pavements, canopies, entrances that ensure protection from rain and sun, fostering continuity, walkability, and a seamless user experience. Uniform design solutions should be made on both sides of the street to create a cohesive streetscape, enhance the public realm, and improve the overall sense of safety.

Retail entrances should be aligned with pavement levels, eliminating steps to ensure accessibility and inclusiveness to everyone.

2. Energy Efficiency Measures

The concept integrates energy-saving strategies to create a sustainable and nearly zero-energy district:

Building Retrofitting: Energy retrofitting for existing buildings to improve thermal performance.

Minergie Certification: Ensuring all buildings meet Swiss Minergie energy efficiency standards.

Renewable Energy Production:

Solar panels integrated into building roofs and facades.

Heat pump systems for district heating.

Localized renewable energy hubs for electricity generation and storage.

Electric Vehicle Infrastructure: Installation of charging stations to support clean mobility.

Bioclimatic Design: Passive solar design, optimizing building orientation.

Use of natural ventilation through courtyard layouts.

Locally sourced materials to reduce embodied energy.

3. Mobility Measures

To enhance connectivity and support sustainable transportation, the proposal includes:

- Development of integrated pedestrian and cycling networks connected to public transport hubs.
- Introduction of car-free zones in key areas to prioritize active mobility.
- Encouraging shared mobility solutions, such as car-sharing and bike-sharing programs.

4. Local Economy Development

The concept supports economic resilience by:

Encouraging small businesses and local entrepreneurship.

- Designing spaces along the streets in Sévelin area for pop-up markets, cultural events, and co-working hubs.
- Supporting retail diversity to meet the needs of a varied demographic.

5. Community Development

Social cohesion is a cornerstone of the proposal, achieved through:

- Development of multi-purpose community centers for cultural and educational activities.
- Inclusive housing strategies targeting diverse age groups and socio-economic demographics.
- Spaces for intergenerational interaction, such as shared gardens and play areas.

6. Public Space Development

Public spaces are reimagined as resilient, inclusive, and climate-adaptive:

Microclimate modeling - Thermal Comfort Measures:

Shading structures, tree planting, and use of reflective materials to mitigate heat, water bodies.

Water Management:

Integration of biotope rainwater collection systems and permeable pavements.
Use of water tanks and separate collectors for irrigation and cleaning.

Green Spaces:

Pocket parks and green corridors to enhance biodiversity.

7. Housing Development

To address urban densification while maintaining quality of life:

- Focus on compact middle-rise housing typologies.
- Integration of at least 10% affordable housing units in any new development in the area or within mixed-use developments.
- Adaptive reuse of industrial buildings (part of Hall Sebeillon) for residential purposes, preserving the area’s historical identity.

Co-benefits expected

Beyond achieving the primary goals of sustainability and urban revitalization, the project is expected to deliver numerous co-benefits:

Enhanced Quality of Life: Residents and visitors will enjoy improved accessibility, greener public spaces, and better thermal comfort, creating a welcoming and enjoyable environment for all.

Social Cohesion: The inclusive design fosters intergenerational and intercultural interactions, strengthening community ties and reducing social disparities.

Economic Resilience: By supporting local businesses, promoting small-scale entrepreneurship, and enhancing retail success through street-level design, the project contributes to a dynamic and resilient local economy.

Environmental Sustainability: Reduced resource consumption, development of soft mobility, renewable energy integration, and efficient mobility systems will contribute to Lausanne’s climate goals and improve the city’s ecological footprint.

Improved Public Health: Safer, pedestrian-friendly streets and accessible public spaces encourage active lifestyles, while air quality improvements from reduced vehicle use and renewable energy sources further enhance well-being.

Principal actions for Integrated redevelopment:

1. New Functional Zoning

(SDG 11 "Sustainable Cities and Communities", SDG 8 "Decent Work and Economic Growth")

2. Street Retrofit

(SDG 8 "Decent Work and Economic Growth", SDG 11 "Sustainable Cities and Communities")

3. Energy Retrofit

(SDG 7 "Affordable and Clean Energy")

4. Densification

(SDG 11 "Sustainable Cities and Communities")

5. Green spaces corridors

(SDG 11 "Sustainable Cities and Communities", SDG 3 "Good Health and Well-Being", SDG 15 "Life on land")

Tools

- **Social Design**
- **Architectural&Engineering Design**
- **Urban Design**
- **Economic design**
- **Governance & Policy**

Figure 42.

Tools (measures):	Aims:	Stakeholders
Social Design	social cohesion	communes
• create community spaces		
• placemaking		
Architectural&Engineering Design	energy optimization	developers, construction companies, municipality, electricity companies
• adaptive re-use (industrial buildings)		
• retrofit of buildings according to new legislation		
• bioclimatic principles for new buildings		
• increase density of buildings		
• implement design codes		
• HVAC system optimization		
• smart meters		
• heat pumps		
• district heating		
• solar panels installations		
Urban Design	social cohesion, economic sustainability, environmental protection	developers, construction companies, municipality, design studios, locals
• street design codes		
• new sustainable public spaces		
• permeable surfaces		
Economic design		locals, businesses, municipaluty
• retail streets		
• create business incubators		
Governance & Policy		municipality, commune

Tools (measures):	Aims:	Stakeholders
• establish participatory planning processes		
• ensure inclusivity		
• local government		
• monitoring: annual KPI reports, feedback loops		

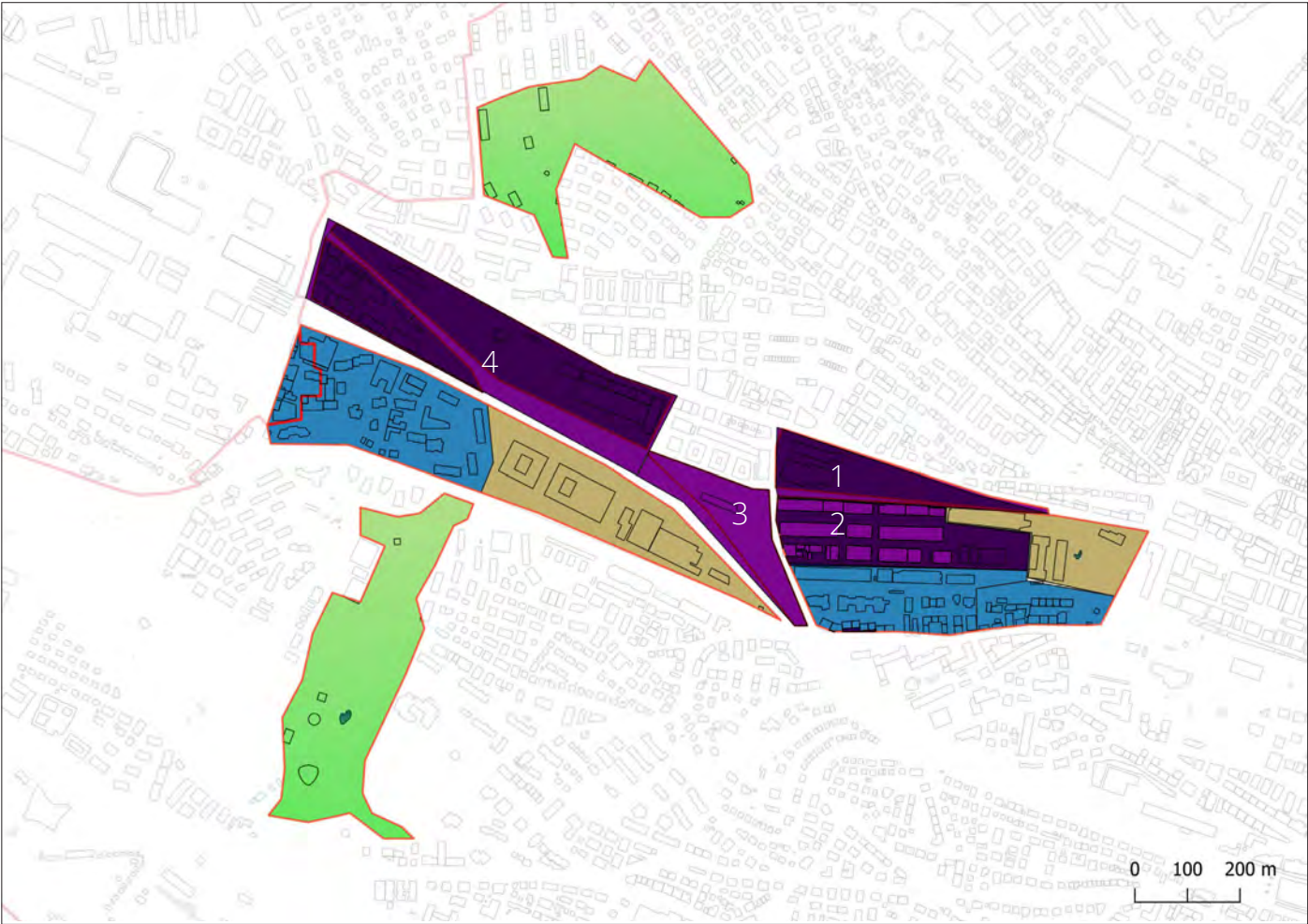


Brainstorm | In the search of ideas and solutions

Credits: authors

1. Concept | New functional zoning

Figure 43. Functional Zoning Proposal 1.



Source: authors

- Boundaries of study
- buildings
- zoning proposal — polygons
- Administrative
- Mixed-use
- Residential
- Parks

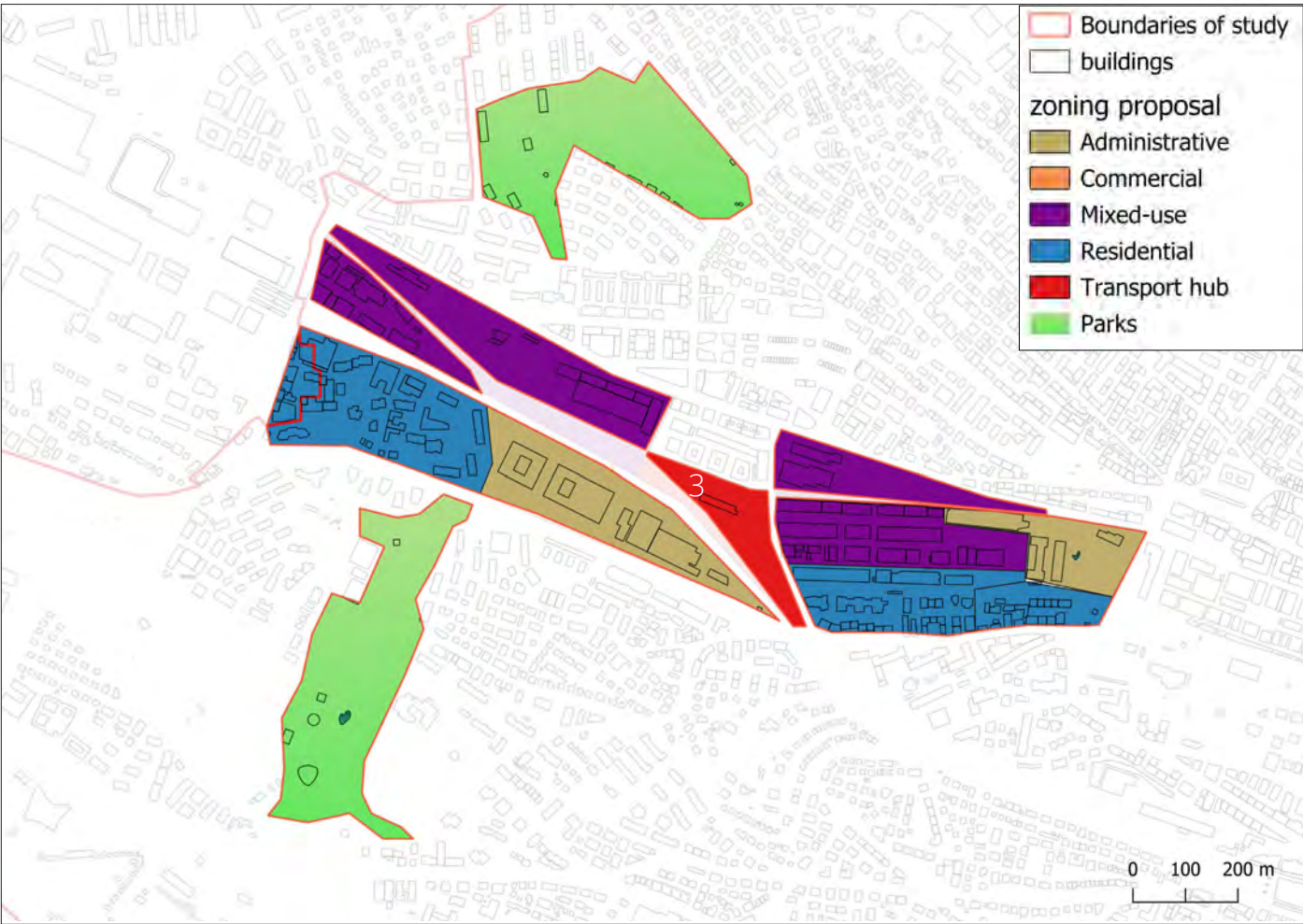
The current functional proposal includes modifications to the existing zones, aiming to facilitate key transformations. At present, residential zones account for only 26.2% of the area. Under the proposed zoning changes, residential areas will be expanded through the establishment and densification of Zones 1, 2, and 3, particularly in the Hall Sébeillon area and the warehouses in the eastern part.

These changes will result in a mixed-use zoning strategy, including residential spaces, covering approximately 19.2 hectares or 41% of the total area. This expansion also opens the possibility for the allocation of new affordable housing, potentially improving the KPI for affordable housing from the current 1% to 30%.

Additionally, the introduction of mixed-use zoning will support the development of new local economies. This transformation is expected to promote economic activity, with local economic contributions potentially increasing to 40%.

1. Concept | New functional zoning

Figure 44. Functional Zoning Proposal 2.



Source: authors

Alternatively, if financial feasibility and strategic interest are confirmed by the Municipality, Zone 3 could be developed into a transport hub with the potential for local renewable energy production, utilizing either solar or geothermal energy. As a benchmark, the solution implemented at King’s Cross Railway Station can serve as a point of reference. Additionally, a striking architectural design for this transport hub could establish a strong identity for the area, enhancing recognition and promoting Lausanne as a vibrant tourist destination.

KPI GROWTH EXPECTATION IN

- Affordable housing (allocation of 30%_
- Local economy (growth to 40%)
- Walkability growth
- (up to 80/100)

Co-benefits

- improved social cohesion
- decrease of housing shortage
- boost of local economy
- local energy production

Measures/tools

Areas: Policy and Governance/ Architectural Design/Economic Design

- Focus on compact middle-rise housing typologies.
- Integration of at least 10% affordable housing units in any new development in each area within mixed-use developments.
- Adaptive reuse of industrial buildings (part of Hall Sebeillon) for residential purposes, preserving the area’s historical identity.

Areas: Architectural Design

The proposal prioritizes a mixed-use composition that integrates residential, commercial, cultural, and recreational functions to ensure a dynamic and balanced urban environment.

Inclusive housing strategies targeting diverse age groups and socio-economic demographics: housing layouts that match different needs in the same building.

Costs/benefits

costs:

- legislation procedures
- taking out businesses

benefits

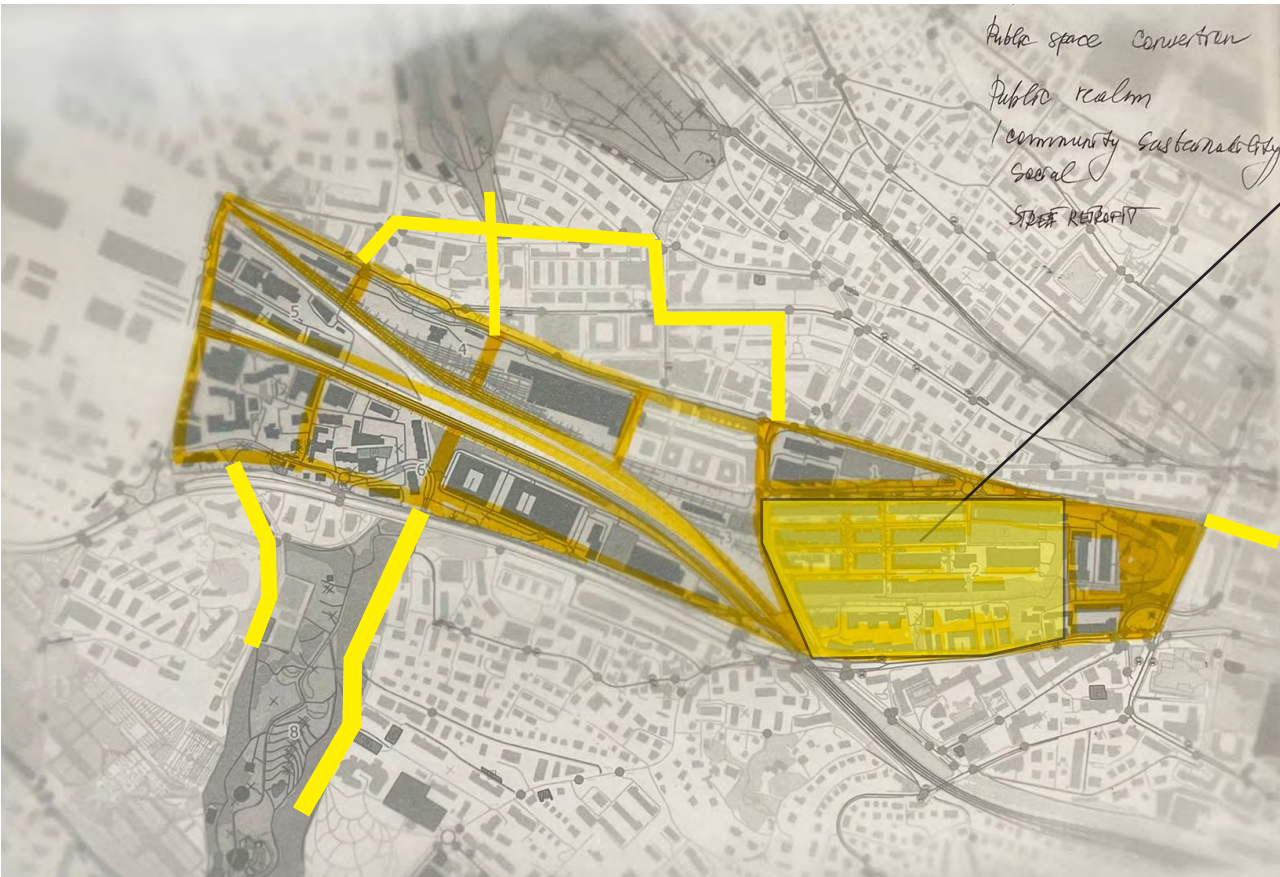
- economic prosperity
- social prosperity and inclusion
- energy saving objectives

Stakeholders involved

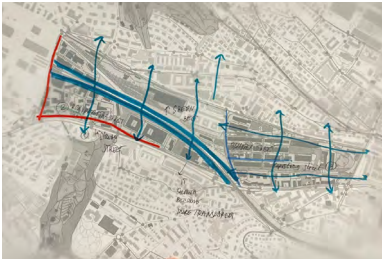
- community
- municipality
- investors
- developers

2. Concept | Street Retrofit | New Connectivity Grid

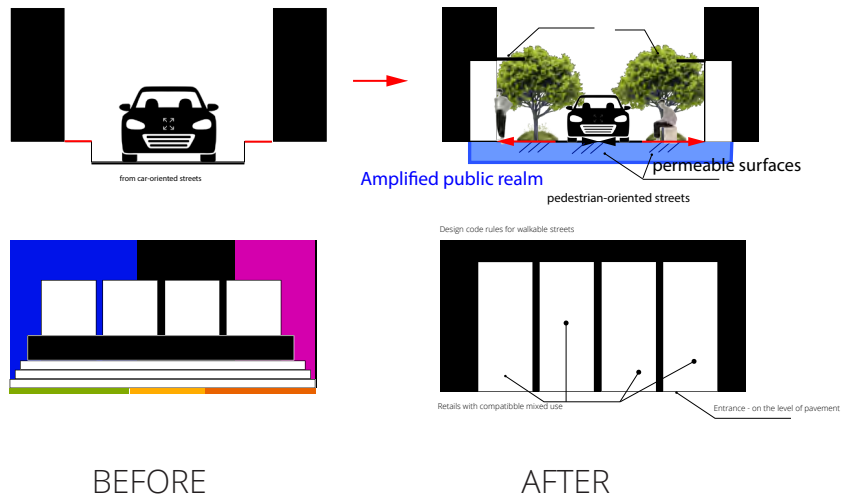
Figure 45. Functional Zoning Proposal 2.



Source: authors



Street retrofit solutions



Measures/tools

Establishment of connectivity with other areas by introducing the same design rules, a street thoroughfare that will create Public Realm and Community Involvement.

Mobility | Areas: Urban Design

- Development of new integrated pedestrian and cycling networks connected to public transport hubs that will increase transparency of the areas.
- The establishment of connectivity with surrounding areas can be achieved by applying consistent design rules, creating a street thoroughfare that forms a green corridor through integrated landscaping.

Local Economy Development | Areas: Urban Design, Economic Design

- Encourage small businesses and local entrepreneurship to settle on new streets and car-free zones.
- Design retail spaces along the ground floor of the newly established pedestrian network
- Support retail diversity to meet the needs of a varied demographic.

Community Development | Areas: Social Design

Development of multi-purpose community centers for cultural and educational activities on the streets
Spaces for intergenerational interaction, such as shared gardens and play areas.
Kids-friendly public spaces
All social groups friendly public spaces

Public Space Development | Areas: Urban Design, Sustainable Landscape Design

Microclimate comfort
Shading structures, tree planting, and use of reflective materials to mitigate heat.
Flood Management:
Integration of biotopes and natural pervious rainwater collection systems (see Image Street Retrofit Solutions) and permeable pavements.
Use of water tanks and separate collectors for irrigation and cleaning.
Green Spaces:
Planting of trees along the streets; new permeable surfaces

Co-benefits

- improved air quality
- improved social cohesion
- boost of local economy
- safety
- flood management
- reduced car dependency

Costs/benefits

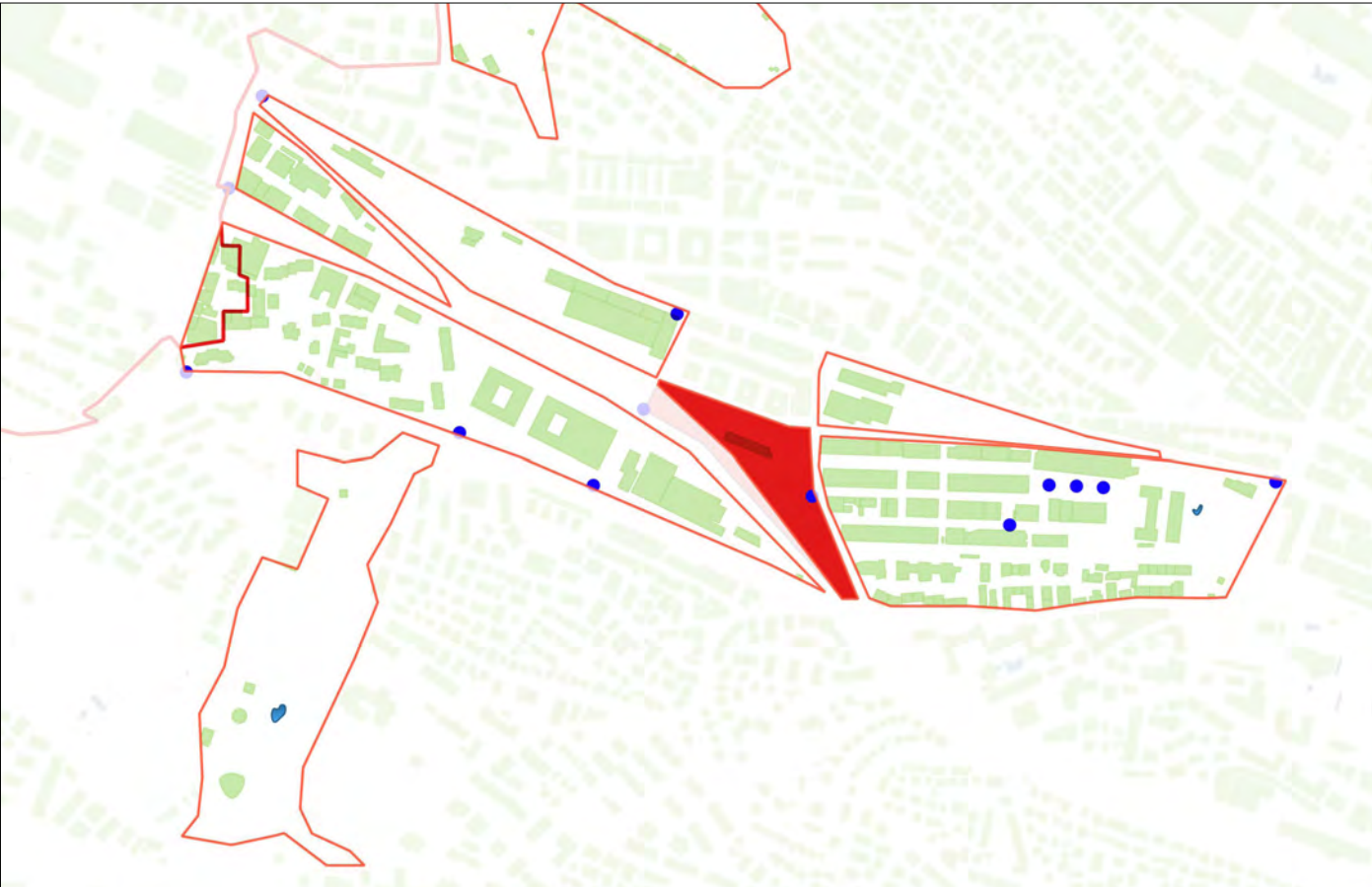
- costs:**
- refurbishment of pavements, streets, retail, design code development
- benefits**
- shared value
 - economic prosperity
 - aesthetic quality
 - safety

Stakeholders involved

- local businesses
- community
- municipality
- design studios

3. Concept | Energy retrofit

Figure 46. Functional Zoning Proposal 2.



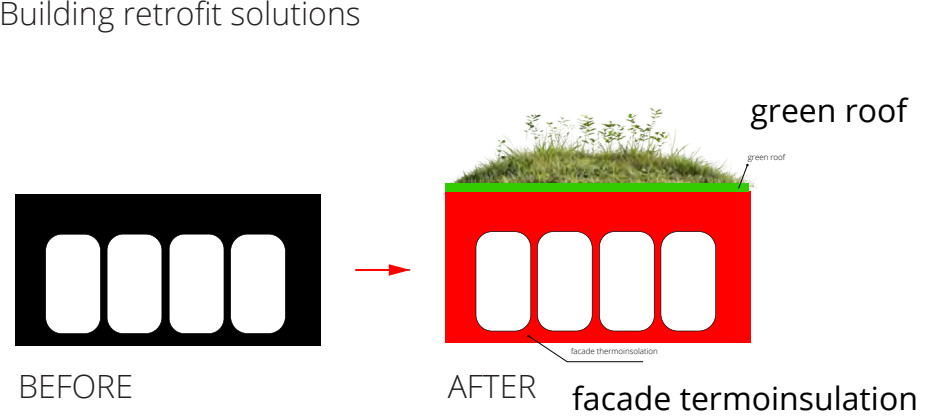
Source: authors

- Boundaries of study
- buildings
- zoning proposal
- Transport hub
- charging stations
- vegetalised roofs

As a proposal it is suggested to create source of local renewable energy production in the neighborhood - an underground geothermal plant analogue to kings cross solutions or a transport hub with big roof surface area (around 9000 m2, analogue to EPFL roofs).

KPI GROWTH EXPECTATION IN

- Energy Consumption per Capita
- Renewable Energy Share



Measures/tools

Areas: Architectural&Engineering Design

The concept integrates energy-saving strategies to create a sustainable and nearly zero-energy district:

- Building Retrofitting:** improve thermal performance.
- Minergie Certification:** Ensuring all buildings meet Swiss Minergie energy efficiency standards.
- Renewable Energy Production:**
 - Solar panels integrated into building roofs and facades.**
 - Heat pump systems for district heating.
 - Localized renewable energy hubs for electricity generation and storage.
- Electric Vehicle Infrastructure:** Installation of charging stations to support clean mobility.
- Bioclimatic Design:** Passive solar design, optimizing building orientation.
- Use of natural ventilation through courtyard layouts.
- Locally sourced materials** to reduce embodied energy.
- Vegetalised Rooftops** to decrease consumption for cooling and heating.
- Diversified local production of energy:** solar, geothermal, etc.

Co-benefits

- improved air quality
- improved energy production and efficiency
- noise reduction

Costs/benefits

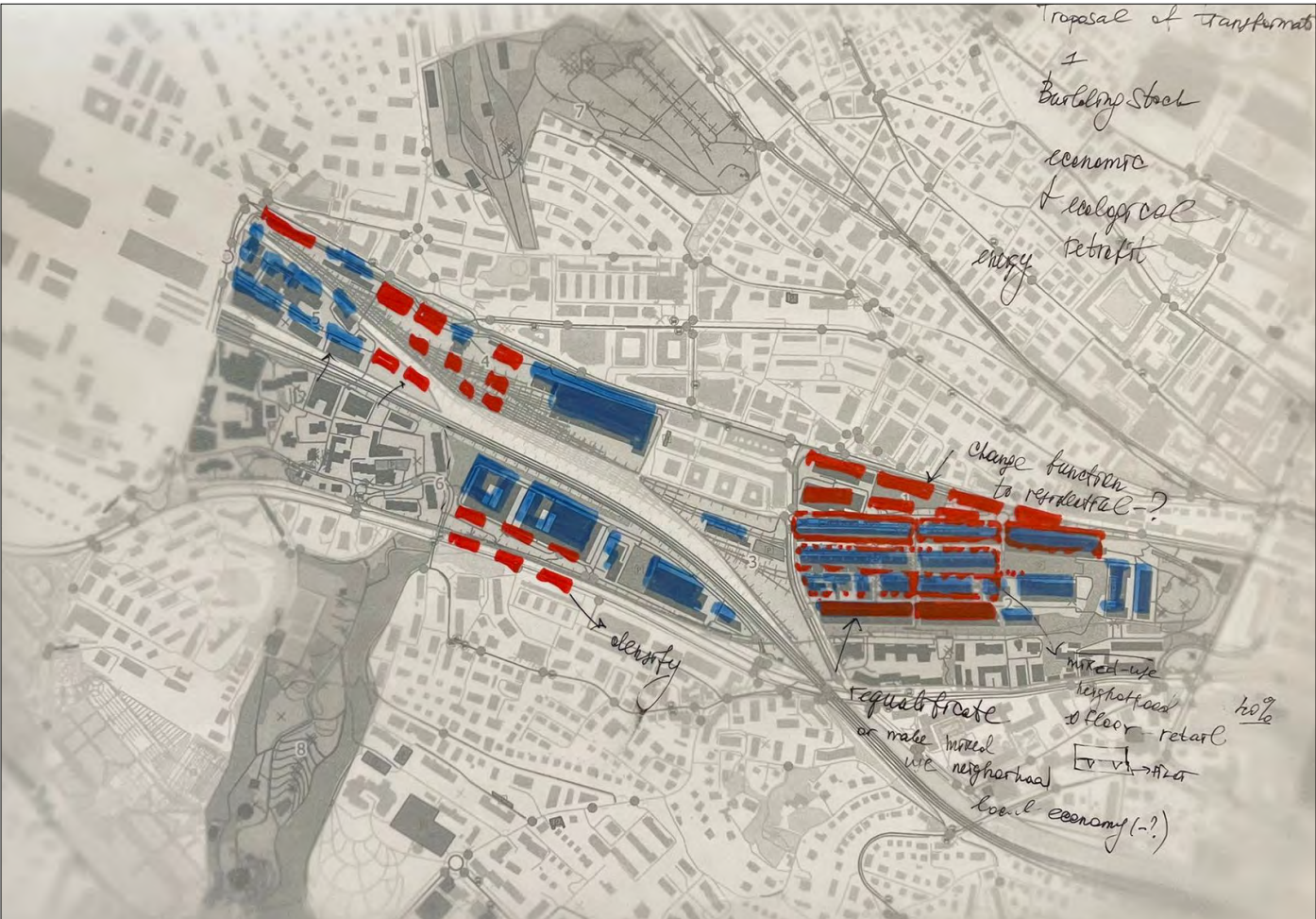
- Costs:
- construction of transport hub and electricity charging facilities
 - refurbishment of roofs
 - technical engineering costs
 - improvement of local population
- Benefits:
- label Energy 2000 kWh society
 - improved local energy production exemplary in nearly zero district

Stakeholders involved

- municipality
- municipality companies
- investors
- developers
- population

4. Concept| Densityfication of buildings/ Housing

Figure 47. Proposal of Densityfication to create straight lined street



Source: authors

Measures/tools

Areas: Architectural&Engineering Design

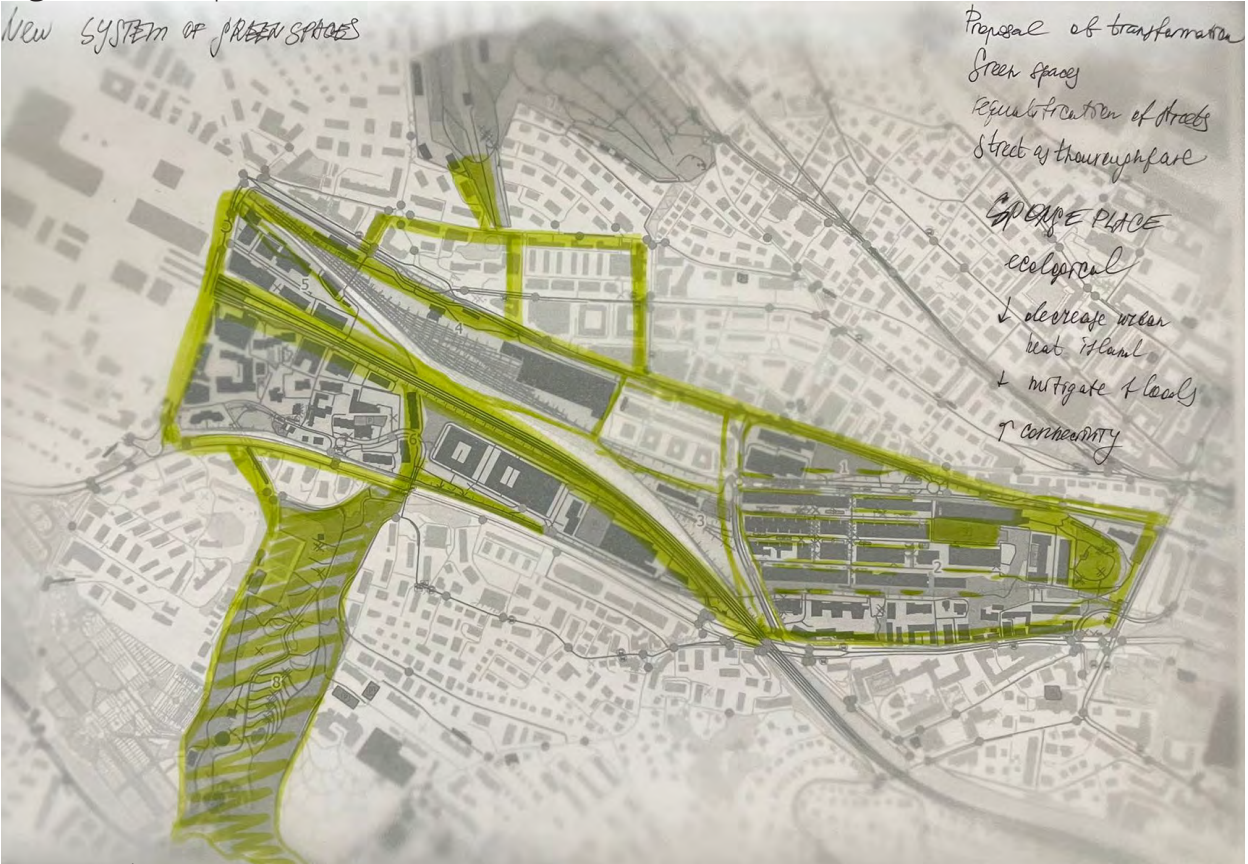
New buildings

The exact number of buildings will be determined by a series of actions that fall outside the scope of this current work. This number will ultimately be defined

	Co-benefits	Costs/benefits	Stakeholders involved
KPI GROWTH EXPECTATION >FAR (2.0-2.5) >% of affordable housing	• housing deficit decrease • sense of safety	costs: taking out businesses construction benefits: housing deficit decrease	• construction companies • population • municipality

5. Concept | Green Spaces Corridors and Parks

Figure 48. Proposal of Green Area Corridors (combined with street retrofit)



Source: authors

KPI GROWTH EXPECTATION

- > Green Area Ratio
- > Gren Spaces Access

Measures/tools

Areas: Urban Design

Development of new integrated pedestrian and cycling networks connected to public transport hubs and increase transparency of the areas

Areas: Urban Design, Sustainable Landscape Design

Thermal Comfort Measures:

Shading structures, tree planting, and use of reflective materials of surfaces to mitigate heat.

Green Spaces:

Pocket parks and green corridors to enhance biodiversity. Design of public plazas for social gatherings and events.

Co-benefits

reduced car dependency (walkability increase)
pollution mitigation
biodiversity
environmental protections
flood risk mitigation
cooling

Costs/benefits

costs:
taking out businesses
material, new construction
benefits
territory attractiveness
cooling

Stakeholders involved

municipality
population
city guests
tourism agencies

Figure 49.
Ecoquartiers criterias

Category	Criteria	Implemented in the proposal
Density, Functional Mix, and Sustainable Mobility	- Dense, mixed-use neighborhoods integrating housing, activities, and services. - Proximity to public transport stops. - Promotion of sustainable mobility through pedestrian and cycling networks. - Safe, pedestrian-friendly environments enhancing user security.	Actions new functional zoning street retrofit energy retrofit densification green spaces corridors
High Environmental Quality	- Reduction of non-renewable resource consumption (soil, energy, water, biodiversity). - Adoption of bioclimatic architectural strategies and efficient technologies. - Use of renewable energies (solar, wood, geothermal, biomass). - Lifecycle analysis and performance monitoring of buildings.	Actions street retrofit energy retrofit green spaces corridors
Intergenerational and Social Mix	- Diverse housing options (size, typology, standards) to cater to a broad audience (students, families, elderly, etc.). - Facilities for intergenerational and sociocultural exchanges (e.g., daycare centers, community spaces, cultural venues, and gardens).	Actions new functional zoning street retrofit densification green spaces corridors
Well-being and Conviviality	- Design of built and unbuilt spaces to enhance quality of life. - Creation of public spaces and amenities (cafés, restaurants, small shops, playgrounds). - Services that foster social interactions and leisure activities.	Actions new functional zoning street retrofit green spaces corridors

Conclusions: The current proposal outlines improvements to the area through several key indicators aligned with ecoquartiers criteria. Ecoquartis is part of Federal Program and is aligned directly with Global Agenda 2030. Alongside proposed improvements, there is a creation of shared value for the entire community.

6 Implementation, monitoring and evaluation

In conclusion, several instruments are suggested to be considered before evaluating the project.
Figure 50.

6.1 Implementation Strategy

Phase	Time-frame	Key Actions	Deliverables
Analysis	0-6 months	Data collection, KPI calculations	Baseline report, GIS maps
Planning	6-12 months	Develop action plan, engage stakeholders	Action plan document, community input
Implementation*	12-36 months	Execute design interventions	Project milestones, quarterly reports
Monitoring*	On-going	Track KPIs, adjust actions as needed	Annual KPI reports, feedback loops

Figure 51.

6.2 Governance & Stakeholder Engagement

Stakeholder	Role	Engagement Method
Local Government	Policy-making, funding	Regular meetings, workshops
Residents	Input on design and priorities	Surveys, public consultations
NGOs	Advocacy, community engagement	Partnerships, workshops
Developers	Investment, construction	PPP agreements
Academic Institutions	Research, innovation	Collaborative projects

7 Conclusions

The current report assessed the state-of-the-art policies in Lausanne, focusing on energy (SDG NN), housing (SDG NN), and public space (SDG NN), showcasing Lausanne as one of the pioneering cities moving toward an energy transition. Additionally, the report provided a detailed analysis of how Switzerland, and particularly Lausanne, aligns local actions with global agendas, addressing climate change and the need for energy consumption optimization.

Using the Malley-Sébeillon neighborhoods as a case study, the report demonstrated how global energy transition strategies (a top-down approach) can and should be balanced with local social inclusion strategies to support and even enhance quality of life and public space. However, a critical question arises: while Lausanne is implementing cutting-edge energy transition technologies and strategies, is this model replicable in other contexts? Exploring this question remains for further discussions.

The report also attempted to develop a tailored system of KPIs, selecting those most appropriate for specific situations within given socio-economic contexts. Although the practice of applying KPIs provides a good orientation for pathways and strategies, the application process and the qualitative selection of indicators is a very subtle question and sometimes have limitations. Methodologically, the work could have been enhanced by still focusing more on local-oriented decisions, such as mechanisms for implementing energy community development and fostering social cohesion. There is always a need to verify proposed solutions with financial feasibility studies, financial Cost/Benefit analysis or Multicriteria Analysis.

Despite these limitations, this exercise in KPI development offers valuable insights. It can serve as a foundation for financial evaluations of urban interventions, providing a holistic view of how one measure can create shared value for the entire community and its individual stakeholders.

Achieving this requires a highly participatory approach, starting as early as the master planning phase. Malley-Sébeillon, despite its current state as a rather neglected and uninviting area, has immense potential. Its redevelopment could elevate Lausanne's reputation, positioning the city as a leader in innovative social transformation.

Contributions of the International Team

This project was a collaborative effort made possible by the diverse expertise and dedication of an international team, each member bringing unique skills to ensure a comprehensive and impactful analysis of the Malley-Sébeillon area.

Florence (Belgian National, Innovation, Human Development and Sustainability): Florence was an invaluable asset as she provided local insights and context. She played a dual role as both a report developer and an end-user, ensuring the project was grounded in Lausanne's realities.

Florence advocated for Malley-Sébeillon as the study area, thoroughly verifying its suitability. She also undertook an in-depth review of local policies and aligned national and regional politics with SDGs and Lausanne's policies. Additionally, she was a peer reviewer for all written texts.

Noelle (Swedish National, Specialist in Sustainable Communities): Noelle brought a deep understanding of sustainability topics and proposed using a KPI system adapted from UNECE. She meticulously selected KPIs, conducted research on target values, and identified best practices for sustainability assessments. Noelle was responsible for analyzing the

alignment of the local agenda with global goals, contributing significantly to the global agenda section. She accompanied all stages of the report, from conception to completion, leaving her mark throughout. Brooke (U.S. National, Specialist in Sustainable Communities):

Brooke contributed as a proofreader and reviser of English texts and played a role in organizing group work. She actively participated in discussions and debates, contributing to the depth of the report. Brooke was responsible for the City Policy Revision, a central part of the report, which she developed with great thoroughness and attention to detail.

Nadezhda (Russian and Brazilian National, Architect and Urban Planning Specialist) Nadezhda played a key role in organizing the work and demonstrated great enthusiasm for engaging with the Lausanne context. She conducted a field study and contributed valuable insights from its outcomes to the report. As a GIS specialist and expert in data analysis, she brought technical expertise to the project. Additionally, her practical experience in urban planning enabled her to contribute creatively to the development of the report. Nadezhda also served as a peer reviewer.

Together, the team seamlessly integrated policy-driven insights and spatial analysis to create a robust framework for sustainable urban transformation. Florence, Noëlle, and Brooke focused on contextual and policy revisions, while Nadya's GIS expertise ensured the redevelopment strategies were technically sound and spatially informed. Their combined efforts ensured that policy frameworks and spatial planning were aligned, resulting in a well-rounded set of recommendations.

This document reflects the collective work of the team and is undersigned by Brooke, Noëlle, Florence, and Nadya.

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